Semantic Matching in Medieval Latin: A Benchmark, Layerwise Analysis, and Geometric Repair

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Abstract

Medieval canon law survives through manuscript families shaped by spelling variation, fragmentary copying, and editorial change. Scholars need strict semantic matching to consolidate duplicate passages and link related witnesses. We introduce a Medieval Latin benchmark with 1,705 canon-law fragments grouped into 840 meaning-equivalent families, and we evaluate pairwise duplicate detection plus open-set directory routing on six pre-trained encoders from three architecture families. A layerwise analysis separates two effects that are easy to conflate. Anisotropy is universal: every model has a layer where the mean cosine between arbitrary passages reaches 0.23 to 0.95, and no baseline layer in any model separates equivalent from non-equivalent passages by more than 0.29 in mean cosine. Catastrophic mid-depth collapse is not universal: the Latin and multilingual T5 encoders fall to AUROC 0.497, 0.539, and 0.654 in their middle layers, while LaBSE, Qwen3-0.6B, and KaLM-mini improve close to monotonically with depth. All-but-the-Top post-processing, fit only on training embeddings, repairs all six models. Best-layer AUROC moves into a 0.971 to 0.984 band, the cosine gap rises to between 0.525 and 0.610, and five-seed top-1 routing accuracy tightens from a 36.6-point spread across models to a 1.0-point spread. SIF pooling supplies a weaker frequency-based correction that helps only the collapsed T5 encoders. We then evaluate local attributions for the same retrieval score with integrated gradients and retrieval-adapted MaRC. The strongest attribution result is narrow: ABTT improves leave-one-out rank faithfulness across all LaTa, PhilTa, and mT5-base model-method cells, while ERASER-style rationale metrics show mixed effects.

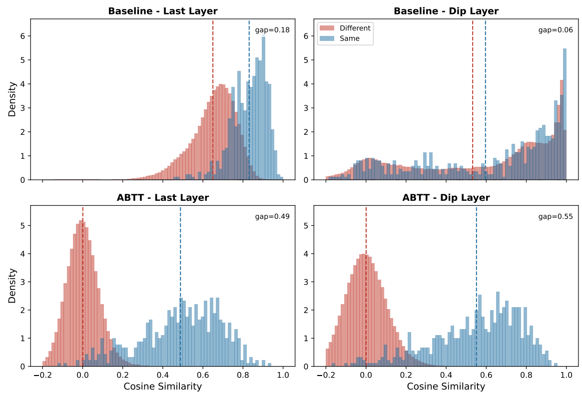

Figure 1: Cosine similarity distributions for PhilTa at the last layer and at the diagnostic mid layer, before and after ABTT. At the collapsed layer, ABTT separates equivalent and non-equivalent passages more clearly.

1 Introduction

Medieval canon law preserves early records of legal reasoning in Western Europe. The corpus does not survive as one stable text. Manuscript copies differ in spelling, wording, fragment boundaries, and editorial history. Scholars often need to decide whether two passages express the same legal content despite those surface changes. This decision supports duplicate consolidation, witness linking, and study of textual transmission.

We study this problem as strict semantic matching. The benchmark contains 1,705 Medieval Latin canon-law fragments arranged into 840 families of meaning-equivalent variants. We evaluate two tasks. Task A asks whether two passages express the same meaning. Task B turns the same embeddings into an open-set workflow: a system either routes a new passage to an existing directory or marks the passage as new.

A simple retrieval system embeds each passage and ranks candidates by cosine similarity. This setup leaves a key modeling issue open. A transformer has many internal layers, and a final-layer sentence embedding does not reveal whether useful retrieval signal appears, disappears, or collapses inside the network. We inspect layers because the failure mode is geometric.

Our layerwise view of six encoders separates two effects that are often reported together. The first is anisotropy, and it is universal. Every model we test has a layer where pooled passages crowd into a narrow cone: the mean cosine between arbitrary passages reaches 0.23 to 0.95, and no baseline layer of any model separates equivalent from non-equivalent passages by more than 0.29 in mean cosine. The second is a catastrophic mid-depth collapse of retrieval, and it is not universal. LaTa, PhilTa, and mT5-base lose almost all pairwise separation in their middle layers, while LaBSE, Qwen3-0.6B, and KaLM-mini improve close to monotonically with depth. The collapse is therefore a symptom of the T5 encoder family on this corpus, not a property of pre-trained encoders in general.

Anisotropy motivates our use of All-but-the-Top (ABTT) post-processing (Mu and Viswanath, 2018). ABTT is not a new method. Here, ABTT serves as a train-free repair after pooling: fit the dominant principal components on training embeddings, remove those directions from train and test embeddings, then score passages with cosine similarity. The repair, unlike the collapse, is universal. ABTT lifts every model into a narrow high-accuracy band on both tasks, and at each T5 encoder's worst baseline layer it recovers pairwise AUROC to within 1.0 point of that same model's best ABTT layer. We also report SIF weighting (Arora et al., 2017), which reweights tokens by frequency before pooling. SIF corrects part of the same dominant direction, but it only helps the models whose retrieval has collapsed, and it never matches ABTT.

Geometry diagnoses collapse and recovery, but it does not select the best retrieval layer. For operational retrieval and attribution we use a train-only rule, described in Section 4.2, which locks LaTa layer 7, PhilTa layer 1, and mT5-base layer 1 before attribution metrics are computed.

We also evaluate local explanations for the retrieval score. Integrated gradients (Sundararajan et al., 2017) gives a gradient view of token importance, and retrieval-adapted MaRC (Brinner and Zarrieß, 2023) gives a learned-mask view after replacing MaRC's classification target with pairwise retrieval cosine. Both explain a fixed query-candidate score, and both remain local diagnostics rather than global accounts of model behavior.

This paper makes four contributions: (1) we introduce a benchmark for strict semantic matching in Medieval Latin canon law, with pairwise and open-set retrieval tasks; (2) we separate universal anisotropy from the T5-specific mid-depth retrieval collapse in a layerwise analysis of six encoders, and we measure both with top-PC dominance, effective rank, and cosine concentration; (3) we show that ABTT repairs all six models without updating parameters, compressing a 36.6-point spread in five-seed top-1 routing accuracy to 1.0 point, and that SIF gives a weaker frequency-based correction that helps only the collapsed models; and (4) we adapt MaRC-style mask optimization to bi-encoder retrieval and compare the method with integrated gradients at predeclared retrieval-selected layers, finding leave-one-out rank faithfulness gains and mixed ERASER-style rationale metrics.

2 Related Work

Anisotropy, layer geometry, and repair. Pre-trained embedding spaces occupy narrow cones (Gao et al., 2019; Ethayarajh, 2019; Godey et al., 2024), where a few coordinates dominate cosine similarity (Kovaleva et al., 2021; Timkey and van Schijndel, 2021). Post-hoc corrections remove or rescale those directions (Mu and Viswanath, 2018; Su et al., 2021) or downweight frequent tokens before pooling (Arora et al., 2017). Probing finds information spread across depth (Tenney et al., 2019; Liu et al., 2019), and intermediate layers can beat the last (Skean et al., 2025). Multilingual analyses are rarer (Rajaee and Pilehvar, 2022). We reuse ABTT unchanged on low-resource retrieval, using geometry to measure anisotropy and a train-only rule to pick layers.

Latin and classical-language NLP. Latin has dedicated encoders and encoder-decoders (Bamman and Burns, 2020; Riemenschneider and Frank, 2023a), and appears in multilingual sentence encoders (Feng et al., 2022), mT5 (Xue et al., 2021), and decoder-only LLM embedders (Zhang et al., 2025; Hu et al., 2025). Shared tasks target tagging (Sprugnoli et al., 2020), retrieval work centres on allusion (Riemenschneider and Frank, 2023b), and charter studies on semantic change (Liu et al., 2024). No general embedding benchmark (Thakur et al., 2021; Muennighoff et al., 2023) covers medieval Latin retrieval.

Text reuse in digital humanities. Text-reuse systems match n-grams (Coffee et al., 2012), align reprinted newspapers (Smith et al., 2013), and add distributional semantics for allusive reuse (Manjavacas et al., 2019; Burns et al., 2021), ranking candidates in a closed collection. Our routing task is open-set: a fragment may belong to no existing family.

Faithfulness of token attributions. Integrated gradients (Sundararajan et al., 2017) and learned input masks (Brinner and Zarrieß, 2023) attribute a score to tokens, and much work asks whether attributions are faithful (Jain and Wallace, 2019; Jacovi and Goldberg, 2020). Erasure and perturbation protocols (DeYoung et al., 2020; Atanasova et al., 2020) target classification; retrieval explanation targets ranking (Anand et al., 2022). We apply them to a bi-encoder cosine score and report where faithfulness improves and where it does not.

3 Task and Dataset

3.1 Task Definition

We study two related tasks.

Task A: pairwise duplicate detection. Let p_1 and p_2 be two passages and let $\equiv$ denote semantic equivalence. The label is

$$y = \begin{cases} 1 & \text{if } p_1 \equiv p_2, \\ 0 & \text{otherwise.} \end{cases}$$

The model produces $\cos(e_{p_1}, e_{p_2})$ from pooled embeddings. Task A uses no directory structure and no threshold at scoring time. Evaluation uses the test $n \times n$ cosine matrix.

Task B: directory routing. Task B uses the same embeddings in a scholar-facing retrieval workflow. Given a query passage q and the existing labelled directories, the system routes q to a directory or marks q as new. We evaluate two routing variants. In autonomous routing, each query receives the maximum cosine score to any other test file. A threshold τ, fit on training same-directory and different-directory pairs, decides whether the query enters its nearest neighbour's directory or the new class. In top-k ranking, each directory receives the best document match score, and a pseudo-option `__NEW__` receives score τ. The simulated expert succeeds when the labelled directory appears in the shortlist.

3.2 Dataset Construction

The corpus contains 1,705 Medieval Latin manuscript fragments from canon law. The labelled files belong to 840 directories, where each directory represents one legal text and its manuscript copies. Directory sizes range from one to ten files. The benchmark asks for strict equivalence, so variants inside a directory count as positive matches and files from different directories count as negatives.

Category	Dirs	Train	Test
Singletons (1 file)	545	272	273
Doubletons (2 files)	108	108	108
Multi-file ($\geq$3)	187	467	477
Total	840	**847**	**858**

Table 1: Train and test split statistics under the v2 varied-allocation protocol. Doubleton directories are assigned as whole folders. The test set contains 535 existing files, with a same-directory partner in test, and 323 new files.

3.3 Train/Test Split

We split the corpus into 847 training files and 858 test files with a leak-free protocol (Table 1).[1] The 545 singleton directories are distributed at the file level. Doubleton directories are assigned as whole folders, 54 to train and 54 to test, so both files from a train doubleton contribute a positive training pair. For directories with at least three files, the split varies the train allocation inside each size class while keeping each class near a balanced file split. This avoids a rule where every size-three directory contributes no positive training pairs. The final split contains 565 positive training pairs and 595 positive test pairs.

All fitting uses training data only. This includes ABTT principal components, routing thresholds, ABTT dimensionality choices, and the operational layer rule used for attribution.

4 Methodology

4.1 Models

We evaluate six pre-trained models from three architecture families:

- **LaTa**: a T5-based seq2seq model pre-trained on Latin (Riemenschneider and Frank, 2023a). We extract encoder hidden states from 12 transformer blocks.
- **PhilTa**: a philologically oriented Latin T5 variant (Riemenschneider and Frank, 2023a), again using 12 encoder blocks.

[1] Implemented in `src/canon_split_v2.py`; deterministic with seed 42.

- **LaBSE**: a language-agnostic BERT sentence encoder covering 109 languages (Feng et al., 2022), with 12 encoder layers.

- **mT5-base**: a multilingual T5 encoder-decoder (Xue et al., 2021), with 12 encoder layers.

- **Qwen3-0.6B**: a multilingual decoder-only embedding model with 28 layers (Zhang et al., 2025).

- **KaLM-mini**: a multilingual decoder-only embedding model with 24 layers (Hu et al., 2025).

4.2 Embedding Extraction and Layer Selection

We extract hidden-state representations from every transformer block, starting at layer 1 and skipping layer 0, the static embedding table. Token representations at each layer are pooled into one passage vector by plain mean pooling or SIF weighting (Arora et al., 2017). For decoder-only models, last-token pooling is another common option; Appendix C compares last-token pooling with mean pooling under ABTT.

The paper-facing pooling baseline removes only tokenizer artifacts with no lexical content, such as whitespace, padding, and tokenizer-prefix-only tokens. Punctuation and numerals remain in the pooled representations. Attribution figures display original decoded tokens, even when a token contributes little to the pooled representation.

Layer geometry and layer selection serve different roles. We use label-free geometry diagnostics, including top-PC dominance, effective rank, and cosine concentration, to identify collapsed layers and test whether ABTT repairs them. For operational retrieval and attribution, we then apply a predeclared train-only retrieval rule. The rule chooses the earliest layer within 0.5 percentage points of the best training-set directory accuracy at rank 1 under ABTT. The rule uses no test attribution outcome. This rule selects LaTa layer 7, PhilTa layer 1, and mT5-base layer 1 for attribution, with $D = 10$ ABTT components. The strongest diagnostic collapse layers are LaTa layer 8, PhilTa layer 6, and mT5-base layer 5, and we keep those as mechanism checks rather than attribution layers.

4.3 Post-Processing

Our main post-processing method is All-but-the-Top (ABTT) (Mu and Viswanath, 2018). For each model and layer, we center the pooled training embeddings, compute the top D principal components by SVD, and remove those components from both train and test embeddings. We search $D \in \{1, 2, 3, 5, 7, 10\}$ and choose D by training-set Task B directory accuracy at rank 1. ABTT itself is borrowed from prior work; our use case is a retrieval-specific repair for anisotropic Latin manuscript representations.

The main comparison also includes SIF weighting (Arora et al., 2017), which divides each token weight by its estimated corpus frequency and so down-weights high-frequency tokens before the passage vector is formed. SIF and ABTT act on the same problem at different stages. SIF reweights tokens before pooling; ABTT removes dominant directions after pooling. Token probabilities for SIF come from training files only. We report four settings throughout: baseline mean pooling, SIF alone, ABTT alone, and SIF followed by ABTT. As a third alternative, PCA whitening (Su et al., 2021) rescales principal directions rather than only removing the top components. It leaves the different-directory cosine mean at or below 0.014, so the learned threshold degenerates: across all 100 whitening cells every file is routed to an existing directory, existing accuracy is 1.000, new accuracy never exceeds one file in 323, and assignment accuracy sits at the 62.4 percent class prior. That prior beats baseline in 60 of 100 cells while directory accuracy at rank 1 never exceeds 0.415, so the apparent gain is a thresholding artifact and we exclude whitening from the main tables.

4.4 Local Attribution Methods

We evaluate local explanations for a fixed query-candidate retrieval score. Integrated gradients (Sundararajan et al., 2017) gives a gradient-based attribution of pairwise cosine similarity to input tokens. We compute per-sequence token attributions for each side of the pair. The derived heatmap in Figure 5 combines token-token cosine similarity with the marginal integrated-gradient scores from the two sequences.

Our learned-mask attribution is adapted from MaRC (Brinner and Zarrieß, 2023). MaRC was proposed for classification, where a soft input mask preserves a target class score while sparsity and smoothness regularizers encourage concise rationales. We keep the per-instance mask-optimization idea and replace the classification target with pairwise retrieval cosine. Given a query-candidate pair, the mask is optimized on one side while the partner representation stays fixed. Candidate-side attribution is obtained by swapping the two sides.

Let $\mathbf{E}(\mathbf{x}_q) \in \mathbb{R}^{T_q \times d_{\text{emb}}}$ denote the input-embedding lookup, $E_{>0}$ the encoder with the lookup removed, and $\mathbf{z}_c^\star$ the candidate representation held fixed during optimization. We parameterize $\mathbf{m} = \sigma(\boldsymbol{\zeta})$ with free logits $\boldsymbol{\zeta} \in \mathbb{R}^{T_q}$ and write $\mathcal{I}$ for content-token positions. For the ABTT variant, define $s_{\text{ABTT}}(\mathbf{a}, \mathbf{b}) = \cos(\text{ABTT}(\mathbf{a}), \mathbf{b})$. The retrieval-adapted MaRC loss is

$$\begin{aligned}\mathcal{L}(\mathbf{m}) = &\big(s_{\text{ABTT}}(\tilde{\mathbf{z}}_q(\mathbf{m}), \mathbf{z}_c^\star) - s_{\text{ABTT}}(\mathbf{z}_q, \mathbf{z}_c^\star)\big)^2 \\ &+ \lambda \sum_{i \in \mathcal{I}} \frac{m_i}{|\mathcal{I}|} + \gamma \sum_{i, i+1 \in \mathcal{I}} \frac{|m_{i+1} - m_i|}{|\mathcal{I}| - 1},\end{aligned} \tag{1}$$

where $\tilde{\mathbf{z}}_q(\mathbf{m}) = \text{pool}(E_{>0}(\text{diag}(\mathbf{m})\, \mathbf{E}(\mathbf{x}_q)), \mathbf{a}_q)$. The baseline variant uses raw cosine in place of the ABTT-corrected cosine. We run 200 Adam steps per pair, initialize $\zeta_0 = \log 9$, and freeze logits outside $\mathcal{I}$ at $-\infty$.

4.5 Evaluation Metrics

Task A. Both Task A metrics come from the test $n \times n$ pairwise cosine matrix. AUROC treats same-directory test pairs as positives and different-directory pairs as negatives. Cosine gap is the mean same-directory cosine minus the mean different-directory cosine.

Task B. For autonomous routing, we report overall assignment accuracy, existing accuracy, and new accuracy. Assignment accuracy scores only the existing-versus-new decision: a file with a partner in test is correct when its maximum cosine to another test file is at least τ, and a file without one is correct when that maximum falls below τ. It never checks which directory the file would enter, so a degenerate threshold that routes everything as existing already attains the 62.4 percent class prior. Read it next to directory accuracy at rank 1, which does require the correct directory. For top-k ranking, cumulative top-K accuracy is the fraction of test queries whose labelled directory appears among the top K options.

Attribution. For attribution, we adapt ERASER-style sufficiency and comprehensiveness metrics (DeYoung et al., 2020) to retrieval cosine. Sufficiency keeps the top-ranked query tokens and measures how much of the full pair cosine remains. Comprehensiveness removes those tokens and measures the score drop. MinFrac reports the smallest retained query-token fraction needed to recover a fixed share of the full cosine. We also report ρ_{LOO}, the Spearman correlation between attribution magnitude and the leave-one-out drop in the same retrieval score.

5 Results

The results move from diagnosis to use: Task A per model and correction, then the layerwise view separating universal anisotropy from the T5-specific collapse, then Task B routing and the train-only layer rule, then attribution at those locked layers.

5.1 Task A: Pairwise Duplicate Detection

Table 2 gives the best-layer Task A result for every model under the four post-processing settings. Two patterns matter. First, baseline quality is uneven: AUROC ranges from 0.839 for mT5-base to 0.970 for KaLM-mini. Second, ABTT removes most of that unevenness. Every model lands between 0.971 and 0.984, and the cosine gap rises from a baseline range of 0.056 to 0.238 up to a range of 0.525 to 0.610. The gain is largest where the baseline is weakest: mT5-base gains 0.140 AUROC, LaTa 0.032, and KaLM-mini 0.011.

The cosine gap column shows why. AUROC and gap disagree: KaLM-mini has the best baseline AUROC at 0.970 but the smallest gap at 0.056, so its ranking is right while its score carries almost no margin for a threshold. ABTT raises every model's gap by at least 0.28, which is what makes a single learned threshold usable in Task B.

SIF as a frequency-based complement. SIF and ABTT act on the same dominant direction from different sides, and Tables 2 and 3 show how far the frequency-based side gets. On Task B assignment accuracy, SIF alone adds 9.4 points for LaTa, 8.2 for PhilTa, and 7.3 for mT5-base, the three models whose retrieval collapses. It adds 0.1 point for LaBSE and subtracts 1.9 for KaLM-mini and 6.3 for Qwen3-0.6B. This fits the mechanism: a common component built mostly from frequent tokens can be attenuated by reweighting those tokens before pooling, but one that survives that reweighting needs the geometric step after pooling. SIF alone never matches ABTT on either task for any model,

	Task A AUROC				Task A cosine gap			
Model	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	0.938_{12}	0.944_{1}	$\mathbf{0.971}_{12}$	0.965_{7}	0.238_{12}	0.365_{1}	0.525_{12}	$\mathbf{0.579}_{7}$
PhilTa	0.939_{1}	0.953_{1}	0.976_{1}	$\mathbf{0.981}_{8}$	0.231_{1}	0.319_{1}	$\mathbf{0.573}_{1}$	0.573_{8}
mT5-base	0.839_{12}	0.846_{1}	$\mathbf{0.979}_{1}$	0.971_{1}	0.082_{12}	0.036_{1}	$\mathbf{0.550}_{1}$	0.519_{1}
LaBSE	0.957_{12}	0.956_{12}	$\mathbf{0.984}_{11}$	0.974_{10}	0.183_{12}	0.184_{12}	$\mathbf{0.610}_{11}$	0.579_{10}
Qwen3-0.6B	0.958_{28}	0.941_{22}	0.972_{2}	$\mathbf{0.972}_{2}$	0.098_{28}	0.053_{22}	$\mathbf{0.553}_{2}$	0.540_{2}
KaLM-mini	0.970_{23}	0.963_{23}	$\mathbf{0.980}_{1}$	0.974_{4}	0.056_{23}	0.056_{23}	$\mathbf{0.568}_{1}$	0.535_{4}

Table 2: Task A pairwise duplicate detection for all six models under four post-processing settings. For every model and method we select one layer by training-set AUROC and report the test-set numbers at that layer; the subscript gives the selected layer. ABTT uses D tuned on train. Baseline AUROC spans 0.839 to 0.970 across the six models, and ABTT lifts every model into a 0.971 to 0.984 band. Cosine gap is the mean same-directory cosine minus the mean different-directory cosine. Per-layer grids are in Appendix Tables 7 and 11, and the SIF sweeps in Appendix Table 14.

and adding SIF before ABTT moves best-layer assignment accuracy by at most 1.8 points either way. We read SIF as a partial account of the anisotropy rather than a competing repair, and keep the full sweeps in Appendix E.

5.2 Layerwise Geometry Diagnosis

Figures 2 and 3 show the layerwise retrieval profile, and they separate the two effects. The retrieval collapse belongs to the T5 family. Baseline AUROC bottoms out at 0.497 for LaTa at layer 6, which is chance, at 0.539 for PhilTa at layer 10, and at 0.654 for mT5-base at layer 5. ABTT at those same layers gives 0.963, 0.982, and 0.977. The other three behave differently: baseline AUROC rises with depth, from 0.807 to 0.957 for LaBSE, from 0.859 to 0.958 for Qwen3-0.6B, and from 0.888 to 0.954 for KaLM-mini, and their layer minima are 0.807, 0.859, and 0.862.

Anisotropy, in contrast, is present in all six. At each model's most collapsed layer, meaning the layer with the largest top-PC variance share, the mean pairwise cosine on the test split is 0.230 for LaTa, 0.584 for PhilTa, 0.314 for mT5-base, 0.878 for LaBSE, 0.953 for Qwen3-0.6B, and 0.887 for KaLM-mini. ABTT with $D = 10$ drives all six to within 0.002 of zero and raises effective rank from as low as 1.00 to between 147.50 and 176.36 (Appendix A). The two families arrive there by different routes. The T5 encoders concentrate variance in one direction, with top-PC shares of 0.956, 0.862, and 1.000, so the cone is low-rank and retrieval dies with it. The sentence-embedding models hold top-PC shares of 0.506, 0.167, and 0.396 but still carry a large common component, so passages sit close together without losing their relative order. ABTT removes that common component in both cases, which is why the repair generalizes even though the symptom does not.

Figure 1 gives the same diagnosis at the score-distribution level. At PhilTa's collapsed layer, positive and negative cosine scores overlap heavily before ABTT. After ABTT, the distributions separate more clearly.

Two-dimensional projections give a visual check on the same change. Under baseline mean pooling, points from different directories overlap; under pure ABTT with $D = 10$, same-directory points form tighter groups, with winnable-fragment silhouette scores rising by 0.09 to 0.75 on the three headline models. Appendix H shows t-SNE (van der Maaten and Hinton, 2008) and UMAP (McInnes et al., 2018) projections for all six models.

5.3 Task B: Directory Routing

Task B connects the layerwise analysis to the retrieval workflow. We evaluate autonomous routing, where a learned threshold decides between an existing directory and the new class, and top-k ranking, where the system presents directory options to a simulated expert. The full per-layer routing and ranking tables for the headline models are in Appendix B; Appendix D reports the remaining models.

Table 3 reports best-layer autonomous routing. Directory accuracy at rank 1, the column that requires the correct directory, spans 39.3 points under the baseline, from 46.6 for mT5-base to 85.9 for KaLM-mini, so the encoder choice dominates. ABTT compresses that to 3.3 points, from 86.1 to 89.4. Assignment accuracy tracks it, moving from a 40.3-point spread to the range 88.5 to 91.7, with every ABTT cell above every baseline cell.

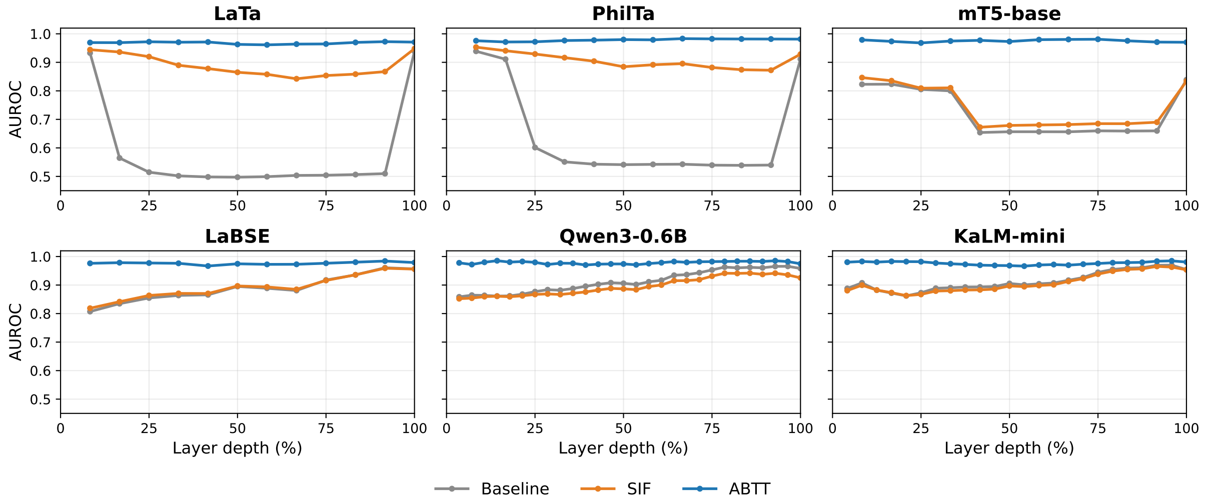

Figure 2: Task A AUROC against normalized layer depth for all six models. The three T5 encoders in the top row lose almost all pairwise separation in their middle layers, with LaTa reaching chance. The three sentence-embedding models in the bottom row improve close to monotonically with depth and never dip below 0.80. ABTT holds every model near its ceiling at every depth, so the repair does not depend on the collapse being present.

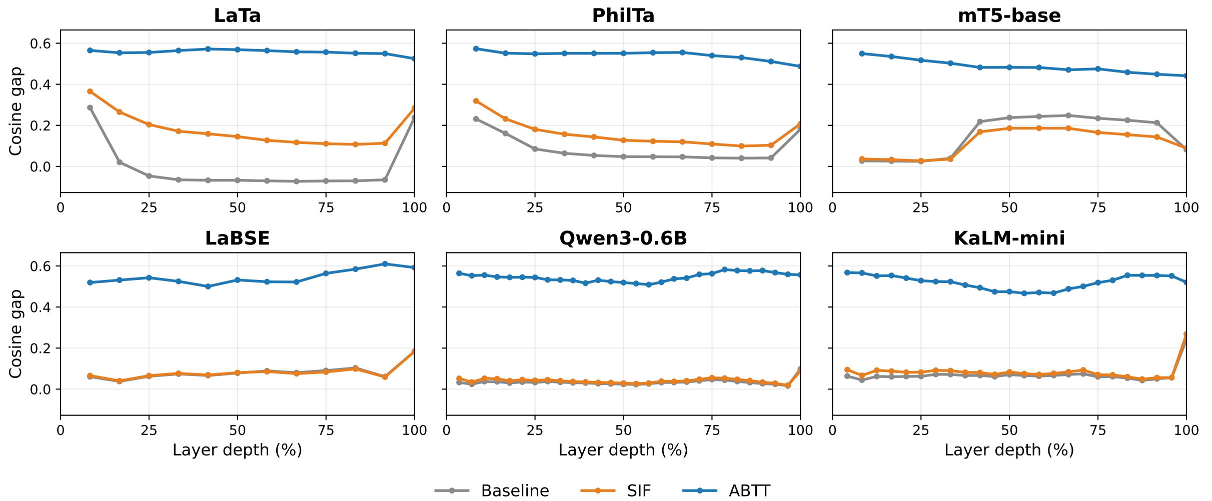

Figure 3: Cosine gap against normalized layer depth for all six models. The baseline gap stays under 0.29 at every layer of every model, and under 0.11 at every layer of Qwen3-0.6B, so raw cosine carries almost no decision margin anywhere. ABTT lifts all six models to a gap near 0.5 across the whole depth profile. SIF raises the gap for LaTa and PhilTa and leaves the remaining four models close to baseline.

Table 4 repeats the top-1 comparison under five query and reference reseedings, so the compression is not an artifact of one draw. The baseline spread of 36.6 points, from 50.6 for mT5-base to 87.3 for KaLM-mini, falls to 1.0 point under the best per-model SIF plus ABTT configuration, from 89.4 to 90.4, with standard deviations of at most 1.1 points.

The same training signal also selects attribution layers. We choose the earliest layer within 0.5 percentage points of the best training-set directory accuracy at rank 1 under ABTT. This rule selects LaTa layer 7, PhilTa layer 1, and mT5-base layer 1. The choice is locked before attribution metrics are computed, so the attribution results do not choose their own layer.

The cumulative view also describes the scholar-facing shortlist. By Top-2, every model exceeds 95%. Gains after Top-3 are small, so two or three options cover nearly all successful cases in this split.

5.4 Attribution at Retrieval-Selected Layers

Table 5 reports the main attribution comparison, and Figure 4 in Appendix F plots the same rank-

	Assignment accuracy				Directory accuracy @1			
Model	Base	SIF	ABTT	SIF+ABTT	Base	SIF	ABTT	SIF+ABTT
LaTa	73.8$_{1}$	83.2$_{1}$	88.5$_{8}$	**90.2**$_{12}$	72.1$_{1}$	81.8$_{1}$	86.1$_{8}$	**88.2**$_{12}$
PhilTa	70.9$_{1}$	79.0$_{1}$	91.1$_{1}$	**91.4**$_{1}$	69.3$_{1}$	77.7$_{1}$	88.3$_{1}$	**88.9**$_{1}$
mT5-base	47.2$_{12}$	54.5$_{1}$	**90.3**$_{1}$	**90.3**$_{2}$	46.6$_{12}$	54.1$_{1}$	**88.5**$_{1}$	87.5$_{2}$
LaBSE	83.3$_{11}$	83.4$_{12}$	**90.8**$_{11}$	89.6$_{11}$	81.5$_{11}$	81.9$_{12}$	**88.5**$_{11}$	86.9$_{11}$
Qwen3-0.6B	82.3$_{28}$	76.0$_{27}$	**91.5**$_{5}$	90.8$_{5}$	80.3$_{28}$	74.2$_{27}$	**89.4**$_{5}$	88.6$_{5}$
KaLM-mini	87.5$_{23}$	85.7$_{22}$	**91.7**$_{3}$	90.8$_{1}$	85.9$_{23}$	84.0$_{22}$	**89.4**$_{3}$	88.0$_{1}$

Table 3: Task B autonomous routing for all six models under four post-processing settings, in percent. For every model and method we select one layer by training-set directory accuracy at rank 1 and report the test-set numbers at that layer; the subscript gives the selected layer. Assignment accuracy scores the existing-versus-new decision alone, comparing each file's maximum cosine against the train-fit threshold τ without checking which directory it would enter; a degenerate threshold that routes everything as existing already attains the 62.4 percent class prior. Directory accuracy at rank 1 is the metric that requires the correct directory, so the two columns should be read together. Per-layer grids are in Appendix Tables 8 and 12, and the SIF sweeps in Appendix Table 15.

Model	**Top-1**	**Top-2**	**Top-3**	**Top-4**	**Top-5**
LaTa	90.1 ± 1.1	96.0 ± 0.4	97.1 ± 0.1	97.6 ± 0.4	97.7 ± 0.4
PhilTa	90.0 ± 0.9	95.7 ± 0.4	96.9 ± 0.4	97.4 ± 0.3	97.7 ± 0.2
LaBSE	90.4 ± 0.8	95.5 ± 0.4	96.4 ± 0.3	97.1 ± 0.5	97.7 ± 0.4
Qwen3-0.6B	90.3 ± 0.6	95.5 ± 0.5	96.8 ± 0.3	97.2 ± 0.5	97.4 ± 0.4
KaLM-mini	89.4 ± 0.5	95.1 ± 0.4	96.4 ± 0.2	96.9 ± 0.3	97.3 ± 0.4
mT5-base	89.8 ± 0.4	95.6 ± 0.5	96.6 ± 0.3	96.9 ± 0.2	97.2 ± 0.2

Table 4: Cumulative top-K accuracy for Task B across the six models, as mean $\pm$ standard deviation over five query/reference reseedings at a fixed v2 split, using the best per-model SIF+ABTT configuration. Each cell gives the percentage of test queries whose labelled directory appears within the top K options. Per-layer top-k rankings are in Appendix Tables 9, 13, and 16.

faithfulness column. The experiment uses 200 positive query-candidate pairs per model for LaTa, PhilTa, and mT5-base. Each model has baseline and ABTT variants for both integrated gradients and retrieval-adapted MaRC. The scores being explained are repaired at these layers: LaTa layer 7 moves from AUROC 0.499 to 0.961 and directory accuracy 0.302 to 0.868, PhilTa layer 1 from 0.939 to 0.976 and 0.693 to 0.883, and mT5-base layer 1 from 0.823 to 0.979 and 0.451 to 0.885.

The clearest quantitative result is rank faithfulness. ABTT improves ρ_{LOO} in all six model-method cells: LaTa IG $-0.001 \rightarrow 0.396$, LaTa MaRC $0.042 \rightarrow 0.401$, PhilTa IG $0.182 \rightarrow 0.607$, PhilTa MaRC $0.190 \rightarrow 0.331$, mT5-base IG $0.164 \rightarrow 0.597$, and mT5-base MaRC $0.250 \rightarrow 0.392$. In those cases, tokens with higher attribution align more closely with leave-one-out drops in the same retrieval score.

The ERASER-style metrics qualify this result. At the global headline thresholds, ABTT improves only 6 of 18 Sufficiency, Comprehensiveness, and MinFrac comparisons. MinFrac@0.80 worsens in all six cells. We therefore read the attribution evidence as support for the repair through rank faithfulness, not as a uniform gain across rationale metrics.

Appendix G shows example-level views for one PhilTa retrieval case, including the integrated-gradient pair matrix before and after ABTT.

6 Discussion

The six-model view supports a narrower claim than the usual one. Anisotropy is universal on this corpus and so is the repair, but the dramatic mid-depth collapse is not. Every model has a layer where passages crowd into a cone, and no model produces a cosine margin above 0.29 at any layer without correction. ABTT then moves all six into a 0.971 to 0.984 AUROC band and a 3.3-point routing band. Only the T5 encoders turn their anisotropy into a retrieval collapse. A study of one architecture family could easily report that collapse as the general finding; three families show that the general finding is the geometry and the correction, while the collapse is a family-level symptom.

The practical reading depends on the family. With a T5 encoder the middle layers are unusable without correction and layer choice matters a great deal. With a sentence-embedding model the final

Model	Method	ρ_{LOO} base	ρ_{LOO} ABTT	$\Delta\rho_{\text{LOO}}$	Suff@25% base→ABTT	Comp@25% base→ABTT	MinFrac@0.80 base→ABTT	ERASER wins
LaTa	IG	-0.001	**0.396**	+0.397	**0.976**→0.681	**0.719**→0.581	**0.041**→0.379	0/3
	MaRC	0.042	**0.401**	+0.359	0.534→**0.645**	0.361→**0.539**	**0.383**→0.459	2/3
PhilTa	IG	0.182	**0.607**	+0.425	0.959→**0.963**	**0.969**→0.747	**0.079**→0.141	1/3
	MaRC	0.190	**0.331**	+0.141	**0.911**→0.754	**0.963**→0.521	**0.104**→0.358	0/3
mT5-base	IG	0.164	**0.597**	+0.433	0.892→**1.027**	0.110→**0.717**	**0.116**→0.214	2/3
	MaRC	0.250	**0.392**	+0.142	**0.913**→0.827	0.042→**0.592**	**0.066**→0.346	1/3

Table 5: **Main-text candidate attribution metrics at the predeclared operational layers for LaTa, PhilTa, and mT5-base.** We report 200 positive query-candidate pairs per model and the two attribution methods used in the paper: integrated gradients (IG) and retrieval-adapted MaRC. The primary faithfulness signal is ρ_{LOO}, which improves under ABTT in 6/6 model–method cells. ERASER-style metrics are shown as baseline→ABTT values at one global threshold choice: Sufficiency and Comprehensiveness at 25% of query tokens, and MinFrac at recovery threshold 0.80 (lower is better for MinFrac, higher otherwise). These midpoint thresholds are applied globally rather than tuned per model or method; Appendix F exposes the full sweeps and Figure 4. Boldface inside the ERASER cells marks the better variant, making disagreements visible: ABTT wins 6/18 ERASER-style comparisons even though ρ_{LOO} improves consistently. Cross-variant comparisons are descriptive because ABTT changes both the attribution scores and the cosine function being explained.

layers are already usable, and the correction still buys 3.5 to 9.1 points of directory accuracy at rank 1 for one SVD per layer fit on 847 training files.

Geometry diagnoses the failure mode but does not select layers. A reader should inspect geometry to find collapse and recovery, then choose retrieval layers with a train-only task signal; in our attribution experiment this gives layers different from the strongest diagnostic collapse layers.

The attribution results need a narrow reading. Because ABTT changes the cosine function being explained, cross-variant comparisons are descriptive: the stable result is improved leave-one-out rank faithfulness in all six cells, while ERASER-style metrics show threshold-dependent tradeoffs.

Limitations

The study covers one domain, medieval canon law, and one language, Latin. The 1,705-fragment corpus is large for this field, but small by current NLP standards. We test six models, and models outside this set might behave differently. The mid-depth retrieval collapse we document holds for the three T5 encoders in our set, and our evidence cannot separate the architecture, the pre-training objective, and the particular checkpoints as its cause. ABTT also needs enough text to estimate principal components, so smaller archives need separate study.

Task B simulates a human-in-the-loop workflow by counting a query as successful when its labelled directory appears in the shortlist. This measures ranking quality, not the full judgment load for a working scholar. The attribution analysis is also local to fixed query-candidate scores. This analysis does not give a global explanation of model behavior or replace expert reading of retrieved passages. The consistent attribution effect is improved leave-one-out rank faithfulness; other rationale metrics remain threshold-dependent and should not be reported as a broad attribution-quality gain.

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A Layer Diagnostics and Attribution Layer Selection

This appendix separates diagnostic layers from operational attribution layers. The diagnostic collapse layers identify where geometry is most anisotropic. The operational attribution layers are selected by training-set retrieval accuracy.

Across the main-model layer grid, PC1 dominance and low effective rank are strongly associated with ABTT gains over baseline. These associations explain why ABTT helps, but they do not replace the train-only retrieval rule used for layer selection.

B Expanded Per-Layer Results for Headline Models

The main text compresses the headline retrieval results into Tables 2 and 3, which report one train-selected layer per model and method. This appendix provides the expanded per-layer Task A duplicate-detection, Task B autonomous-routing, and Task B top-k ranking tables for LaTa, PhilTa, and mT5-base.

Model	Operational layer	Diagnostic layer	PC1 before	PC1 after	Eff. rank before→after
LaTa	7	8	0.956	0.036	1.34→168.40
PhilTa	1	6	0.862	0.039	1.83→155.02
mT5-base	1	5	1.000	0.044	1.00→151.74

Table 6: Diagnostic collapse layers and operational attribution layers for the three headline models. PC1 and effective-rank values compare baseline geometry with ABTT-D10 at the diagnostic collapse layer. Operational layers come from the train-only retrieval rule.

Model	Layer	Task A: Pairwise Duplicate Detection (main)			
		AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
LaTa	1	0.933	0.969	0.286	0.565
	2	0.565	0.969	0.020	0.553
	3	0.515	0.972	-0.046	0.555
	4	0.502	0.970	-0.065	0.564
	5	0.498	0.971	-0.068	0.572
	6	0.497	0.963	-0.067	0.569
	7	0.499	0.961	-0.070	0.564
	8	0.504	0.964	-0.073	0.558
	9	0.504	0.964	-0.071	0.557
	10	0.507	0.970	-0.070	0.551
	11	**0.510**	**0.973**	**-0.065**	**0.550**
	12	0.938	0.971	0.238	0.525
PhilTa	1	0.939	0.976	0.231	0.573
	2	0.911	0.971	0.161	0.551
	3	0.601	0.972	0.085	0.549
	4	0.551	0.976	0.064	0.551
	5	0.543	0.978	0.054	0.551
	6	0.541	0.980	0.048	0.551
	7	0.542	0.979	0.047	0.554
	8	**0.543**	**0.983**	**0.047**	**0.555**
	9	0.539	0.982	0.042	0.540
	10	0.539	0.982	0.040	0.530
	11	0.540	0.982	0.041	0.511
	12	0.911	0.981	0.181	0.487
mT5-base	1	0.823	0.979	0.027	0.550
	2	0.823	0.973	0.026	0.535
	3	0.805	0.968	0.025	0.517
	4	0.800	0.975	0.040	0.503
	5	0.654	0.977	0.218	0.482
	6	0.657	0.973	0.238	0.482
	7	0.657	0.979	0.243	0.482
	8	0.656	0.980	0.248	0.471
	9	**0.660**	**0.981**	**0.235**	**0.475**
	10	0.659	0.975	0.225	0.459
	11	0.660	0.971	0.213	0.449
	12	0.839	0.970	0.082	0.441

Table 7: Per-layer Task A pairwise duplicate-detection metrics for the three headline models (LaTa, PhilTa, mT5-base). `baseline` is mean pooling without correction; `abtt_optimal` applies ABTT (no SIF weighting) with D tuned per layer on the train split. Both AUROC and cosine gap are computed over the test $\mathrm{n}\times\mathrm{n}$ cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the best layer per model, selected by ABTT AUROC.

Model	Layer	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
		Task B: Autonomous Routing (main, file-level, τ-thresholded)					
LaTa	**1**	**0.630**	**0.910**	**0.916**	**0.907**	**0.738**	**0.909**
	2	0.153	0.905	0.913	0.892	0.439	0.900
	3	0.256	0.882	0.842	0.950	0.477	0.908
	4	0.389	0.893	0.718	0.920	0.513	0.903
	5	0.374	0.892	0.715	0.913	0.502	0.900
	6	0.378	0.884	0.712	0.913	0.503	0.895
	7	0.372	0.875	0.715	0.916	0.501	0.890
	8	0.379	0.884	0.706	0.885	0.502	0.885
	9	0.366	0.886	0.724	0.873	0.501	0.881
	10	0.344	0.873	0.752	0.882	0.498	0.876
	11	0.310	0.867	0.786	0.901	0.490	0.880
	12	0.576	0.890	0.932	0.833	0.710	0.868
PhilTa	**1**	**0.591**	**0.918**	**0.904**	**0.901**	**0.709**	**0.911**
	2	0.479	0.895	0.867	0.885	0.625	0.892
	3	0.193	0.897	0.889	0.879	0.455	0.890
	4	0.148	0.886	0.923	0.873	0.439	0.881
	5	0.252	0.879	0.796	0.873	0.457	0.876
	6	0.338	0.905	0.700	0.817	0.474	0.872
	7	0.301	0.901	0.746	0.830	0.469	0.874
	8	0.262	0.875	0.808	0.885	0.467	0.879
	9	0.303	0.890	0.762	0.870	0.476	0.882
	10	0.299	0.908	0.780	0.793	0.480	0.865
	11	0.256	0.869	0.824	0.836	0.470	0.857
	12	0.338	0.880	0.898	0.876	0.549	0.879
mT5-base	**1**	**0.135**	**0.877**	**0.981**	**0.947**	**0.453**	**0.903**
	2	0.110	0.890	0.975	0.910	0.436	0.897
	3	0.161	0.794	0.926	0.941	0.449	0.850
	4	0.269	0.787	0.858	0.882	0.491	0.823
	5	0.422	0.746	0.659	0.864	0.512	0.790
	6	0.636	0.755	0.489	0.836	0.580	0.786
	7	0.594	0.793	0.517	0.851	0.565	0.815
	8	0.557	0.809	0.560	0.842	0.558	0.822
	9	0.364	0.841	0.746	0.814	0.508	0.831
	10	0.314	0.794	0.783	0.876	0.491	0.825
	11	0.297	0.763	0.780	0.854	0.479	0.797
	12	0.198	0.834	0.926	0.799	0.472	0.821

Table 8: Per-layer Task B autonomous routing accuracy for the three headline models. `baseline` is mean pooling without correction; `abtt_optimal` applies ABTT (no SIF weighting) with D tuned per layer on the train split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on train via F_1-optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbour's directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the best layer per model, selected by ABTT overall assignment accuracy.

Model	Layer	Task B: Top-K Ranking (main, single-seed v2)					
		Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
LaTa	1	0.721	0.880	0.630	0.910	0.916	0.907
	2	0.383	0.869	0.153	0.905	0.913	0.892
	3	**0.350**	**0.882**	**0.256**	**0.882**	**0.842**	**0.950**
	4	0.303	0.881	0.389	0.893	0.718	0.920
	5	0.297	0.875	0.374	0.892	0.715	0.913
	6	0.297	0.874	0.378	0.884	0.712	0.913
	7	0.302	0.868	0.372	0.875	0.715	0.916
	8	0.302	0.861	0.379	0.884	0.706	0.885
	9	0.309	0.858	0.366	0.886	0.724	0.873
	10	0.317	0.852	0.344	0.873	0.752	0.882
	11	0.331	0.857	0.310	0.867	0.786	0.901
	12	0.689	0.841	0.576	0.890	0.932	0.833
PhilTa	**1**	**0.693**	**0.883**	**0.591**	**0.918**	**0.904**	**0.901**
	2	0.603	0.869	0.479	0.895	0.867	0.885
	3	0.389	0.865	0.193	0.897	0.889	0.879
	4	0.369	0.859	0.148	0.886	0.923	0.873
	5	0.334	0.854	0.252	0.879	0.796	0.873
	6	0.310	0.845	0.338	0.905	0.700	0.817
	7	0.325	0.847	0.301	0.901	0.746	0.830
	8	0.350	0.857	0.262	0.875	0.808	0.885
	9	0.337	0.857	0.303	0.890	0.762	0.870
	10	0.344	0.833	0.299	0.908	0.780	0.793
	11	0.355	0.829	0.256	0.869	0.824	0.836
	12	0.528	0.854	0.338	0.880	0.898	0.876
mT5-base	**1**	**0.451**	**0.885**	**0.135**	**0.877**	**0.981**	**0.947**
	2	0.435	0.874	0.110	0.890	0.975	0.910
	3	0.445	0.830	0.161	0.794	0.926	0.941
	4	0.467	0.798	0.269	0.787	0.858	0.882
	5	0.352	0.756	0.422	0.746	0.659	0.864
	6	0.346	0.746	0.636	0.755	0.489	0.836
	7	0.352	0.775	0.594	0.793	0.517	0.851
	8	0.366	0.784	0.557	0.809	0.560	0.842
	9	0.394	0.787	0.364	0.841	0.746	0.814
	10	0.387	0.788	0.314	0.794	0.783	0.876
	11	0.378	0.765	0.297	0.763	0.780	0.854
	12	0.466	0.779	0.198	0.834	0.926	0.799

Table 9: Per-layer Task B top-k ranking metrics for the three headline models. `baseline` is mean pooling without correction; `abtt_optimal` applies ABTT (no SIF weighting) with D tuned per layer on the train split. This table is computed on the single-seed v2 split (no M-seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 16 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the best layer per model, selected by ABTT Acc@1.

C Last-Token vs. Mean Pooling

Decoder-only embedding models often use last-token pooling. We compare last-token pooling with mean pooling under the same ABTT pipeline, train/test split, and ABTT choices. Table 10 reports the layer with the best Task B autonomous-routing accuracy, Task B directory accuracy at rank 1, and cosine gap at the same layer. Mean pooling performs better on assignment accuracy for every model in this experiment, including the decoder-only models.

D Per-Layer Results for Remaining Models

The main text focuses on LaTa, PhilTa, and mT5-base. This appendix reports the same per-layer view for LaBSE, Qwen3-0.6B, and KaLM-mini under the same comparison: baseline mean pooling versus ABTT with D tuned per layer on the train split.

Table 10: Mean vs. last-token pooling on the balanced split, restricted to ABTT post-processing. Best layer per (model, pooling) selected by Task A assignment accuracy. Task B metrics (Acc@1, cosine gap) reported at that same layer.

Model	Pool	Layer	Method	Assign Acc	Acc@1	Gap
LaTa	Mean	1	ABTT (D=10)	0.909	0.880	0.565
LaTa	LastTok	12	ABTT (D=10)	0.676	0.425	0.276
PhilTa	Mean	1	ABTT (D=10)	0.911	0.883	0.573
PhilTa	LastTok	9	ABTT (D=10)	0.604	0.385	0.200
LaBSE	Mean	11	ABTT (D=10)	0.908	0.885	0.610
LaBSE	LastTok	12	ABTT (D=10)	0.895	0.871	0.586
Qwen3-0.6B	Mean	5	ABTT (D=10)	0.915	0.894	0.545
Qwen3-0.6B	LastTok	27	ABTT (D=10)	0.907	0.885	0.562
KaLM-mini	Mean	1	ABTT (D=10)	0.921	0.902	0.568
KaLM-mini	LastTok	24	ABTT (D=10)	0.875	0.847	0.483
Qwen3-8B	Mean	1	ABTT (D=10)	0.924	0.897	0.543
Qwen3-8B	LastTok	35	ABTT (D=10)	0.910	0.892	0.642

Table 11: Per-layer Task A pairwise metrics for the remaining three models (LaBSE, Qwen3-0.6B, KaLM-mini), under the same two-method comparison as the main paper. `baseline` is mean pooling without correction; `abtt_optimal` applies ABTT (no SIF weighting) with D tuned per layer on the train split. Both AUROC and cosine gap are computed over the test n×n cosine matrix with no threshold and no directory routing: AUROC treats same-directory test pairs as positives, and cosine gap is the difference between mean same-directory and mean different-directory cosine. Rows in bold mark the best layer per model, selected by ABTT AUROC.

		Task A: Pairwise Duplicate Detection (appendix models)			
Model	**Layer**	AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
LaBSE	1	0.807	0.976	0.060	0.519
	2	0.835	0.978	0.038	0.531
	3	0.855	0.977	0.063	0.543
	4	0.864	0.976	0.074	0.525
	5	0.866	0.967	0.066	0.500
	6	0.895	0.974	0.078	0.532
	7	0.889	0.973	0.089	0.523
	8	0.881	0.973	0.080	0.522
	9	0.918	0.976	0.091	0.564
	10	0.936	0.980	0.103	0.584
	11	**0.961**	**0.984**	**0.061**	**0.610**
	12	0.957	0.979	0.183	0.592
Qwen3-0.6B	1	0.859	0.978	0.033	0.564
	2	0.864	0.972	0.024	0.553
	3	0.864	0.980	0.037	0.555
	4	**0.861**	**0.985**	**0.036**	**0.546**
	5	0.862	0.980	0.030	0.545
	6	0.868	0.983	0.035	0.545
	7	0.877	0.980	0.033	0.544
	8	0.884	0.972	0.037	0.533
	9	0.882	0.977	0.034	0.532
	10	0.888	0.976	0.031	0.530
	11	0.896	0.970	0.030	0.516
	12	0.903	0.973	0.027	0.531
	13	0.908	0.974	0.026	0.524
	14	0.906	0.974	0.024	0.519
	15	0.902	0.971	0.022	0.514
	16	0.911	0.975	0.026	0.509
	17	0.917	0.978	0.032	0.521
	18	0.934	0.982	0.032	0.537
	19	0.937	0.980	0.035	0.541
	20	0.943	0.982	0.041	0.559
	21	0.953	0.982	0.047	0.562
	22	0.963	0.983	0.044	0.583
	23	0.960	0.983	0.038	0.578
	24	0.962	0.984	0.032	0.576
	25	0.960	0.983	0.027	0.578
	26	0.965	0.985	0.024	0.568
	27	0.966	0.982	0.017	0.559
	28	0.958	0.975	0.098	0.556
KaLM-mini	1	0.888	0.980	0.064	0.568
	2	0.908	0.983	0.044	0.566
	3	0.882	0.980	0.062	0.552
	4	0.872	0.983	0.060	0.553
	5	0.862	0.982	0.062	0.541
	6	0.873	0.982	0.062	0.528
	7	0.889	0.977	0.072	0.524
	8	0.891	0.975	0.072	0.523
	9	0.893	0.972	0.066	0.507
	10	0.893	0.970	0.067	0.494
	11	0.895	0.969	0.061	0.474
	12	0.905	0.968	0.071	0.475
	13	0.901	0.966	0.066	0.467
	14	0.904	0.970	0.063	0.471
	15	0.907	0.972	0.067	0.468

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Model	Layer	Task A: Pairwise Duplicate Detection (appendix models)			
		AUROC (base)	AUROC (ABTT)	Cosine gap (base)	Cosine gap (ABTT)
	16	0.917	0.970	0.071	0.488
	17	0.926	0.973	0.075	0.501
	18	0.944	0.976	0.060	0.519
	19	0.955	0.978	0.061	0.530
	20	0.959	0.979	0.054	0.554
	21	0.961	0.980	0.043	0.554
	22	0.969	0.984	0.050	0.554
	23	**0.970**	**0.985**	**0.056**	**0.552**
	24	0.954	0.981	0.243	0.520

Table 12: Per-layer Task B autonomous routing accuracy for the remaining three models. `baseline` is mean pooling without correction; `abtt_optimal` applies ABTT (no SIF weighting) with D tuned per layer on the train split. For every test file we compute $s_i = \max_{j \neq i} \cos(e_i, e_j)$ against the rest of the test set and compare to the learned threshold τ (fit on train via F_1-optimal cut on same-directory vs. different-directory pairs). The decision is binary first (existing if $s_i \geq \tau$, else new); only files predicted existing are then routed to their top-1 neighbour's directory. We report **existing accuracy** (restricted to files whose true partner is in the test set), **new accuracy** (files that should be flagged as novel), and **overall assignment accuracy**. Rows in bold mark the best layer per model, selected by ABTT overall assignment accuracy.

Model	Layer	Task B: Autonomous Routing (appendix models)					
		Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
LaBSE	1	0.222	0.879	0.898	0.873	0.477	0.876
	2	0.381	0.875	0.811	0.827	0.543	0.857
	3	0.277	0.886	0.907	0.867	0.514	0.879
	4	0.372	0.897	0.885	0.836	0.565	0.874
	5	0.327	0.845	0.901	0.851	0.543	0.847
	6	0.548	0.860	0.882	0.851	0.674	0.857
	7	0.398	0.849	0.892	0.830	0.584	0.841
	8	0.385	0.869	0.870	0.743	0.568	0.822
	9	0.421	0.871	0.929	0.817	0.612	0.851
	10	0.606	0.892	0.892	0.805	0.713	0.859
	11	**0.785**	**0.923**	**0.913**	**0.882**	**0.833**	**0.908**
	12	0.729	0.914	0.938	0.864	0.808	0.895
Qwen3-0.6B	1	0.264	0.921	0.960	0.885	0.526	0.908
	2	0.437	0.869	0.814	0.916	0.579	0.887
	3	0.250	0.914	0.947	0.907	0.513	0.911
	4	0.411	0.936	0.848	0.842	0.576	0.901
	5	**0.181**	**0.910**	**0.932**	**0.923**	**0.464**	**0.915**
	6	0.422	0.935	0.830	0.796	0.576	0.882
	7	0.250	0.890	0.947	0.876	0.513	0.885
	8	0.409	0.931	0.895	0.799	0.592	0.881
	9	0.460	0.901	0.851	0.851	0.607	0.882
	10	0.533	0.892	0.836	0.864	0.647	0.881
	11	0.293	0.895	0.978	0.814	0.551	0.865
	12	0.407	0.912	0.938	0.817	0.607	0.876
	13	0.432	0.908	0.944	0.836	0.625	0.881
	14	0.456	0.907	0.926	0.811	0.633	0.871
	15	0.469	0.901	0.920	0.845	0.639	0.880
	16	0.639	0.897	0.830	0.808	0.711	0.864
	17	0.480	0.886	0.935	0.851	0.652	0.873
	18	0.553	0.882	0.944	0.839	0.700	0.866
	19	0.514	0.871	0.966	0.858	0.684	0.866
	20	0.654	0.865	0.898	0.892	0.746	0.875
	21	0.647	0.912	0.950	0.898	0.761	0.907
	22	0.731	0.888	0.938	0.916	0.809	0.899
	23	0.622	0.914	0.966	0.895	0.752	0.907
	24	0.748	0.901	0.935	0.935	0.818	0.914
	25	0.574	0.893	0.975	0.923	0.725	0.904

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Model	Layer	Task B: Autonomous Routing (appendix models)					
		Existing (base)	Existing (ABTT)	New (base)	New (ABTT)	Overall (base)	Overall (ABTT)
	26	0.613	0.899	0.975	0.895	0.749	0.897
	27	0.793	0.877	0.833	0.923	0.808	0.894
	28	0.796	0.877	0.867	0.913	0.823	0.890
KaLM-mini	**1**	**0.391**	**0.921**	**0.907**	**0.920**	**0.585**	**0.921**
	2	0.493	0.916	0.848	0.879	0.627	0.902
	3	0.443	0.923	0.870	0.907	0.604	0.917
	4	0.308	0.931	0.935	0.895	0.544	0.917
	5	0.338	0.905	0.882	0.923	0.543	0.911
	6	0.256	0.895	0.910	0.920	0.502	0.904
	7	0.475	0.864	0.854	0.944	0.618	0.894
	8	0.467	0.862	0.870	0.913	0.619	0.881
	9	0.469	0.873	0.885	0.870	0.626	0.872
	10	0.449	0.875	0.885	0.870	0.613	0.873
	11	0.458	0.862	0.885	0.864	0.619	0.862
	12	0.507	0.862	0.895	0.904	0.653	0.878
	13	0.540	0.860	0.895	0.870	0.674	0.864
	14	0.593	0.836	0.845	0.885	0.688	0.854
	15	0.566	0.864	0.898	0.827	0.691	0.850
	16	0.550	0.849	0.935	0.830	0.695	0.841
	17	0.675	0.864	0.920	0.889	0.767	0.873
	18	0.697	0.875	0.960	0.898	0.796	0.883
	19	0.721	0.873	0.966	0.892	0.814	0.880
	20	0.705	0.791	0.966	0.913	0.803	0.837
	21	0.727	0.811	0.966	0.926	0.817	0.854
	22	0.850	0.879	0.916	0.923	0.875	0.895
	23	0.849	0.884	0.920	0.898	0.875	0.889
	24	0.787	0.892	0.944	0.867	0.846	0.882

Table 13: Per-layer Task B top-k ranking metrics for the remaining three models. `baseline` is mean pooling without correction; `abtt_optimal` applies ABTT (no SIF weighting) with D tuned per layer on the train split. This table is computed on the single-seed v2 split (no M-seed averaging, since the mseed sweep was run only for SIF-conditioned variants; see Table 16 for the multi-seed SIF+ABTT view). **Acc@1** is directory accuracy at rank 1, with an existing/new decomposition. Rows in bold mark the best layer per model, selected by ABTT Acc@1.

Model	Layer	Task B: Top-K Ranking (appendix models, single-seed)					
		Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
LaBSE	1	0.465	0.851	0.222	0.879	0.898	0.873
	2	0.494	0.830	0.381	0.875	0.811	0.827
	3	0.500	0.848	0.277	0.886	0.907	0.867
	4	0.542	0.841	0.372	0.897	0.885	0.836
	5	0.521	0.811	0.327	0.845	0.901	0.851
	6	0.640	0.831	0.548	0.860	0.882	0.851
	7	0.561	0.810	0.398	0.849	0.892	0.830
	8	0.544	0.779	0.385	0.869	0.870	0.743
	9	0.608	0.823	0.421	0.871	0.929	0.817
	10	0.693	0.832	0.606	0.892	0.892	0.805
	11	**0.815**	**0.885**	**0.785**	**0.923**	**0.913**	**0.882**
	12	0.791	0.868	0.729	0.914	0.938	0.864
Qwen3-0.6B	1	0.521	0.875	0.264	0.921	0.960	0.885
	2	0.558	0.871	0.437	0.869	0.814	0.916
	3	0.508	0.890	0.250	0.914	0.947	0.907
	4	0.559	0.874	0.411	0.936	0.848	0.842
	5	0.450	0.894	0.181	0.910	0.932	0.923
	6	0.551	0.858	0.422	0.935	0.830	0.796
	7	0.507	0.859	0.250	0.890	0.947	0.876
	8	0.580	0.852	0.409	0.931	0.895	0.799
	9	0.587	0.852	0.460	0.901	0.851	0.851
	10	0.622	0.848	0.533	0.892	0.836	0.864
	11	0.548	0.833	0.293	0.895	0.978	0.814

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Model	Layer	Task B: Top-K Ranking (appendix models, single-seed)					
		Acc@1 (base)	Acc@1 (ABTT)	Existing (base)	Existing (ABTT)	New (base)	New (ABTT)
	12	0.603	0.848	0.407	0.912	0.938	0.817
	13	0.618	0.851	0.432	0.908	0.944	0.836
	14	0.621	0.840	0.456	0.907	0.926	0.811
	15	0.627	0.848	0.469	0.901	0.920	0.845
	16	0.682	0.830	0.639	0.897	0.830	0.808
	17	0.641	0.840	0.480	0.886	0.935	0.851
	18	0.691	0.837	0.553	0.882	0.944	0.839
	19	0.676	0.837	0.514	0.871	0.966	0.858
	20	0.733	0.852	0.654	0.865	0.898	0.892
	21	0.752	0.881	0.647	0.912	0.950	0.898
	22	0.795	0.876	0.731	0.888	0.938	0.916
	23	0.742	0.886	0.622	0.914	0.966	0.895
	24	**0.808**	**0.896**	**0.748**	**0.901**	**0.935**	**0.935**
	25	0.718	0.886	0.574	0.893	0.975	0.923
	26	0.740	0.881	0.613	0.899	0.975	0.895
	27	0.787	0.876	0.793	0.877	0.833	0.923
	28	0.803	0.872	0.796	0.877	0.867	0.913
KaLM-mini	**1**	**0.577**	**0.902**	**0.391**	**0.921**	**0.907**	**0.920**
	2	0.607	0.879	0.493	0.916	0.848	0.879
	3	0.587	0.894	0.443	0.923	0.870	0.907
	4	0.538	0.887	0.308	0.931	0.935	0.895
	5	0.524	0.882	0.338	0.905	0.882	0.923
	6	0.490	0.876	0.256	0.895	0.910	0.920
	7	0.592	0.873	0.475	0.864	0.854	0.944
	8	0.597	0.860	0.467	0.862	0.870	0.913
	9	0.601	0.843	0.469	0.873	0.885	0.870
	10	0.584	0.844	0.449	0.875	0.885	0.870
	11	0.586	0.836	0.458	0.862	0.885	0.864
	12	0.622	0.850	0.507	0.862	0.895	0.904
	13	0.633	0.840	0.540	0.860	0.895	0.870
	14	0.636	0.829	0.593	0.836	0.845	0.885
	15	0.650	0.817	0.566	0.864	0.898	0.827
	16	0.676	0.819	0.550	0.849	0.935	0.830
	17	0.747	0.852	0.675	0.864	0.920	0.889
	18	0.777	0.865	0.697	0.875	0.960	0.898
	19	0.801	0.861	0.721	0.873	0.966	0.892
	20	0.789	0.817	0.705	0.791	0.966	0.913
	21	0.802	0.831	0.727	0.811	0.966	0.926
	22	0.857	0.875	0.850	0.879	0.916	0.923
	23	0.859	0.872	0.849	0.884	0.920	0.898
	24	0.833	0.858	0.787	0.892	0.944	0.867

E SIF-Variant Post-Processing Sweeps

The main text uses pure ABTT as the paper-facing repair. This appendix reports SIF-only, SIF plus fixed-D ABTT, and SIF plus tuned-D ABTT. The tables keep one primary metric per task so the four-method grid stays readable.

Table 14: Per-layer Task A AUROC across the SIF-conditioned post-processing suite for all six models. `sif_only` replaces mean pooling with SIF-weighted pooling; `sif_abtt_fixed` adds ABTT with fixed D=10; `sif_abtt_optimal` tunes D per layer on the train split. Gap is omitted to keep the table narrow; the pure-ABTT comparison (not SIF-conditioned) is in Tables 7 and 11. Rows in bold mark the best layer per model, selected by SIF+ABTT (D-tuned) AUROC.

Model	Layer	Task A: SIF-suite AUROC (appendix)			
		AUROC (base)	AUROC (SIF)	AUROC (SIF+ABTT$_{10}$)	AUROC (SIF+ABTT)
LaTa	**1**	**0.933**	**0.944**	**0.971**	**0.974**
	2	0.565	0.936	0.964	0.970
	3	0.515	0.920	0.966	0.964
	4	0.502	0.890	0.969	0.968
	5	0.498	0.878	0.969	0.969
	6	0.497	0.865	0.964	0.964
	7	0.499	0.858	0.965	0.965
	8	0.504	0.842	0.964	0.964
	9	0.504	0.854	0.964	0.964
	10	0.507	0.859	0.965	0.965
	11	0.510	0.867	0.965	0.965
	12	0.938	0.948	0.963	0.963
PhilTa	1	0.939	0.953	0.973	0.973
	2	0.911	0.941	0.970	0.970
	3	0.601	0.929	0.966	0.966
	4	0.551	0.916	0.970	0.970
	5	0.543	0.904	0.971	0.976
	6	0.541	0.884	0.980	0.980
	7	0.542	0.891	0.980	0.980
	8	**0.543**	**0.895**	**0.981**	**0.981**
	9	0.539	0.882	0.977	0.977
	10	0.539	0.874	0.977	0.977
	11	0.540	0.872	0.978	0.978
	12	0.911	0.929	0.968	0.968
mT5-base	1	0.823	0.846	0.971	0.971
	2	0.823	0.835	0.964	0.964
	3	0.805	0.809	0.956	0.956
	4	0.800	0.810	0.966	0.966
	5	**0.654**	**0.672**	**0.972**	**0.972**
	6	0.657	0.679	0.970	0.970
	7	0.657	0.680	0.966	0.966
	8	0.656	0.682	0.967	0.967
	9	0.660	0.685	0.966	0.966
	10	0.659	0.685	0.964	0.964
	11	0.660	0.690	0.964	0.964
	12	0.839	0.834	0.968	0.968
LaBSE	1	0.807	0.819	0.974	0.974
	2	0.835	0.842	0.974	0.974
	3	0.855	0.864	0.968	0.968
	4	0.864	0.871	0.966	0.966
	5	0.866	0.870	0.963	0.963
	6	0.895	0.897	0.964	0.964
	7	0.889	0.893	0.961	0.961
	8	0.881	0.885	0.966	0.966
	9	0.918	0.916	0.971	0.971
	10	0.936	0.936	0.974	0.974
	11	0.961	0.959	0.980	0.980
	12	**0.957**	**0.956**	**0.978**	**0.982**
Qwen3-0.6B	**1**	**0.859**	**0.852**	**0.982**	**0.980**
	2	0.864	0.854	0.972	0.972
	3	0.864	0.859	0.977	0.975
	4	0.861	0.861	0.976	0.977
	5	0.862	0.859	0.971	0.971
	6	0.868	0.862	0.974	0.974
	7	0.877	0.867	0.971	0.971

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Model	Layer	Task A: SIF-suite AUROC (appendix) AUROC (base)	AUROC (SIF)	AUROC (SIF+ABTT$_{10}$)	AUROC (SIF+ABTT)
	8	0.884	0.869	0.969	0.969
	9	0.882	0.867	0.968	0.968
	10	0.888	0.871	0.966	0.966
	11	0.896	0.876	0.964	0.964
	12	0.903	0.882	0.967	0.967
	13	0.908	0.888	0.967	0.967
	14	0.906	0.887	0.966	0.966
	15	0.902	0.884	0.967	0.967
	16	0.911	0.895	0.965	0.965
	17	0.917	0.900	0.967	0.974
	18	0.934	0.915	0.966	0.972
	19	0.937	0.916	0.966	0.976
	20	0.943	0.919	0.971	0.971
	21	0.953	0.931	0.971	0.971
	22	0.963	0.941	0.973	0.973
	23	0.960	0.941	0.973	0.973
	24	0.962	0.942	0.975	0.973
	25	0.960	0.938	0.978	0.974
	26	0.965	0.941	0.979	0.979
	27	0.966	0.935	0.977	0.977
	28	0.958	0.925	0.967	0.967
KaLM-mini	1	0.888	0.881	0.981	0.979
	2	0.908	0.900	0.981	0.981
	3	0.882	0.882	0.977	0.977
	4	0.872	0.873	0.974	0.974
	5	0.862	0.863	0.971	0.971
	6	0.873	0.867	0.966	0.966
	7	0.889	0.879	0.971	0.971
	8	0.891	0.880	0.971	0.975
	9	0.893	0.883	0.966	0.966
	10	0.893	0.883	0.963	0.963
	11	0.895	0.886	0.968	0.968
	12	0.905	0.897	0.965	0.965
	13	0.901	0.894	0.963	0.963
	14	0.904	0.898	0.968	0.968
	15	0.907	0.901	0.968	0.968
	16	0.917	0.913	0.966	0.966
	17	0.926	0.923	0.968	0.968
	18	0.944	0.938	0.976	0.976
	19	0.955	0.949	0.978	0.978
	20	0.959	0.955	0.978	0.978
	21	0.961	0.957	0.979	0.982
	22	0.969	0.965	0.982	0.982
	23	**0.970**	**0.963**	**0.986**	**0.984**
	24	0.954	0.954	0.975	0.975

Table 15: Per-layer Task B overall assignment accuracy across the SIF-conditioned suite for all six models. Restricted to overall routing accuracy (existing/new decomposition omitted) to keep the table compact; see Tables 8 and 12 for the pure-ABTT pairwise comparison with existing/new broken out. Rows in bold mark the best layer per model, selected by SIF+ABTT (D-tuned) overall assignment accuracy.

Model	Layer	Task B: SIF-suite routing (appendix) Overall (base)	Overall (SIF)	Overall (SIF+ABTT$_{10}$)	Overall (SIF+ABTT)
LaTa	1	0.738	0.832	0.920	0.899
	2	0.439	0.791	0.904	0.900
	3	**0.477**	**0.720**	**0.910**	**0.907**
	4	0.513	0.668	0.900	0.902
	5	0.502	0.648	0.906	0.906
	6	0.503	0.668	0.895	0.895

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Model	Layer	Task B: SIF-suite routing (appendix) Overall (base)	Overall (SIF)	Overall (SIF+ABTT$_{10}$)	Overall (SIF+ABTT)
	7	0.501	0.642	0.903	0.903
	8	0.502	0.614	0.892	0.892
	9	0.501	0.618	0.889	0.889
	10	0.498	0.655	0.890	0.890
	11	0.490	0.621	0.880	0.880
	12	0.710	0.804	0.902	0.902
PhilTa	**1**	**0.709**	**0.790**	**0.914**	**0.914**
	2	0.625	0.738	0.892	0.892
	3	0.455	0.713	0.892	0.892
	4	0.439	0.663	0.886	0.886
	5	0.457	0.626	0.894	0.897
	6	0.474	0.612	0.906	0.906
	7	0.469	0.625	0.906	0.906
	8	0.467	0.614	0.886	0.886
	9	0.476	0.590	0.895	0.895
	10	0.480	0.580	0.888	0.888
	11	0.470	0.565	0.886	0.886
	12	0.549	0.667	0.899	0.899
mT5-base	**1**	**0.453**	**0.545**	**0.907**	**0.907**
	2	0.436	0.537	0.903	0.903
	3	0.449	0.568	0.889	0.889
	4	0.491	0.488	0.857	0.857
	5	0.512	0.483	0.826	0.826
	6	0.580	0.558	0.817	0.817
	7	0.565	0.554	0.832	0.832
	8	0.558	0.548	0.834	0.834
	9	0.508	0.495	0.845	0.845
	10	0.491	0.495	0.832	0.832
	11	0.479	0.484	0.816	0.816
	12	0.472	0.541	0.858	0.858
LaBSE	1	0.477	0.521	0.886	0.886
	2	0.543	0.508	0.875	0.875
	3	0.514	0.577	0.860	0.860
	4	0.565	0.587	0.869	0.869
	5	0.543	0.598	0.857	0.857
	6	0.674	0.648	0.872	0.872
	7	0.584	0.634	0.858	0.858
	8	0.568	0.571	0.854	0.854
	9	0.612	0.613	0.859	0.859
	10	0.713	0.698	0.864	0.864
	11	**0.833**	**0.807**	**0.896**	**0.896**
	12	0.808	0.834	0.890	0.880
Qwen3-0.6B	1	0.526	0.566	0.895	0.909
	2	0.579	0.524	0.894	0.894
	3	0.513	0.537	0.904	0.901
	4	0.576	0.531	0.899	0.904
	5	0.464	0.537	0.908	0.908
	6	0.576	0.526	0.906	0.906
	7	**0.513**	**0.573**	**0.911**	**0.911**
	8	0.592	0.627	0.911	0.911
	9	0.607	0.576	0.910	0.910
	10	0.647	0.599	0.900	0.900
	11	0.551	0.606	0.893	0.893
	12	0.607	0.663	0.900	0.900
	13	0.625	0.675	0.897	0.897
	14	0.633	0.571	0.893	0.893
	15	0.639	0.600	0.893	0.893
	16	0.711	0.679	0.880	0.880
	17	0.652	0.686	0.900	0.876
	18	0.700	0.731	0.899	0.882
	19	0.684	0.700	0.886	0.850
	20	0.746	0.676	0.882	0.882

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Model	Layer	Task B: SIF-suite routing (appendix) Overall (base)	Overall (SIF)	Overall (SIF+ABTT$_{10}$)	Overall (SIF+ABTT)
	21	0.761	0.739	0.892	0.892
	22	0.809	0.719	0.893	0.893
	23	0.752	0.759	0.893	0.893
	24	0.818	0.706	0.900	0.899
	25	0.725	0.748	0.897	0.893
	26	0.749	0.772	0.903	0.903
	27	0.808	0.760	0.899	0.899
	28	0.823	0.774	0.890	0.890
KaLM-mini	**1**	**0.585**	**0.604**	**0.906**	**0.908**
	2	0.627	0.594	0.908	0.908
	3	0.604	0.576	0.907	0.907
	4	0.544	0.577	0.901	0.901
	5	0.543	0.533	0.897	0.897
	6	0.502	0.558	0.900	0.900
	7	0.618	0.593	0.901	0.901
	8	0.619	0.607	0.892	0.878
	9	0.626	0.663	0.876	0.876
	10	0.613	0.603	0.866	0.866
	11	0.619	0.627	0.874	0.874
	12	0.653	0.642	0.874	0.874
	13	0.674	0.666	0.864	0.864
	14	0.688	0.648	0.861	0.861
	15	0.691	0.698	0.866	0.866
	16	0.695	0.738	0.859	0.859
	17	0.767	0.763	0.882	0.882
	18	0.796	0.796	0.879	0.879
	19	0.814	0.804	0.878	0.878
	20	0.803	0.821	0.876	0.876
	21	0.817	0.832	0.874	0.876
	22	0.875	0.857	0.876	0.876
	23	0.875	0.862	0.880	0.881
	24	0.846	0.847	0.869	0.869

Table 16: Per-layer Task B top-k ranking accuracy across all six models, averaged over $M = 5$ query/reference reseedings of the v2 train/test split (mean ± std). `sif_abtt_optimal` applies SIF weighting plus ABTT with D tuned per layer on the train split; the mseed sweep was not run for the pure `abtt_optimal` variant reported in the main paper, so we include this table to give a variance-aware view of the SIF-conditioned best variant. Overall assignment accuracy coincides with Acc@1 in the mseed pipeline, so we report the existing/new decomposition instead. Rows in bold mark the best layer per model, selected by SIF+ABTT Acc@1.

Model	Layer	Task B: Top-K Ranking (appendix, 5-seed mean ± std, SIF+ABTT) Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
LaTa	1	0.731 ± 0.009	0.889 ± 0.009	0.562 ± 0.004	0.842 ± 0.009	0.949 ± 0.011	0.950 ± 0.011
	2	0.428 ± 0.011	0.887 ± 0.005	0.048 ± 0.006	0.840 ± 0.005	0.920 ± 0.008	0.949 ± 0.009
	3	0.392 ± 0.012	0.889 ± 0.008	0.046 ± 0.008	0.845 ± 0.006	0.840 ± 0.020	0.946 ± 0.013
	4	0.330 ± 0.018	0.888 ± 0.008	0.047 ± 0.008	0.858 ± 0.007	0.696 ± 0.030	0.927 ± 0.013
	5	0.326 ± 0.020	0.890 ± 0.008	0.039 ± 0.004	0.856 ± 0.005	0.698 ± 0.028	0.935 ± 0.014
	6	0.321 ± 0.018	0.890 ± 0.008	0.038 ± 0.007	0.861 ± 0.008	0.688 ± 0.026	0.927 ± 0.014
	7	$\mathbf{0.327 \pm 0.020}$	$\mathbf{0.895 \pm 0.006}$	$\mathbf{0.044 \pm 0.007}$	$\mathbf{0.846 \pm 0.006}$	$\mathbf{0.694 \pm 0.029}$	$\mathbf{0.957 \pm 0.010}$
	8	0.325 ± 0.022	0.883 ± 0.007	0.046 ± 0.010	0.832 ± 0.006	0.685 ± 0.032	0.949 ± 0.011
	9	0.331 ± 0.019	0.885 ± 0.008	0.045 ± 0.010	0.846 ± 0.006	0.701 ± 0.024	0.935 ± 0.013
	10	0.346 ± 0.017	0.888 ± 0.002	0.044 ± 0.010	0.844 ± 0.007	0.736 ± 0.022	0.944 ± 0.010
	11	0.360 ± 0.018	0.881 ± 0.006	0.045 ± 0.009	0.860 ± 0.008	0.767 ± 0.028	0.907 ± 0.011
	12	0.707 ± 0.015	0.889 ± 0.005	0.511 ± 0.017	0.834 ± 0.005	0.961 ± 0.008	0.961 ± 0.010
PhilTa	1	0.706 ± 0.016	0.898 ± 0.007	0.532 ± 0.018	0.856 ± 0.009	0.931 ± 0.012	0.953 ± 0.011
	2	0.625 ± 0.018	0.889 ± 0.009	0.408 ± 0.016	0.866 ± 0.009	0.905 ± 0.017	0.919 ± 0.017
	3	0.439 ± 0.013	0.892 ± 0.006	0.071 ± 0.014	0.856 ± 0.008	0.917 ± 0.013	0.939 ± 0.009
	4	0.420 ± 0.016	0.893 ± 0.009	0.027 ± 0.007	0.858 ± 0.006	0.928 ± 0.015	0.938 ± 0.015

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Model	Layer	Task B: Top-K Ranking (appendix, 5-seed mean ± std, SIF+ABTT) Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
	5	0.374 ± 0.015	0.896 ± 0.010	0.042 ± 0.011	0.872 ± 0.007	0.805 ± 0.018	0.925 ± 0.014
	6	0.338 ± 0.015	0.897 ± 0.008	0.059 ± 0.010	0.861 ± 0.007	0.698 ± 0.025	0.945 ± 0.014
	7	**0.361 ± 0.016**	**0.900 ± 0.009**	**0.056 ± 0.010**	**0.859 ± 0.005**	**0.757 ± 0.021**	**0.952 ± 0.016**
	8	0.390 ± 0.016	0.891 ± 0.010	0.058 ± 0.012	0.858 ± 0.007	0.820 ± 0.017	0.933 ± 0.018
	9	0.372 ± 0.015	0.894 ± 0.006	0.066 ± 0.012	0.861 ± 0.006	0.768 ± 0.026	0.936 ± 0.011
	10	0.377 ± 0.013	0.892 ± 0.006	0.066 ± 0.012	0.855 ± 0.003	0.781 ± 0.021	0.941 ± 0.015
	11	0.394 ± 0.013	0.889 ± 0.006	0.058 ± 0.008	0.851 ± 0.008	0.829 ± 0.018	0.937 ± 0.008
	12	0.563 ± 0.010	0.896 ± 0.007	0.279 ± 0.011	0.863 ± 0.009	0.932 ± 0.005	0.938 ± 0.014
mT5-base	**1**	**0.486 ± 0.012**	**0.898 ± 0.004**	**0.100 ± 0.007**	**0.856 ± 0.009**	**0.987 ± 0.007**	**0.953 ± 0.010**
	2	0.475 ± 0.012	0.892 ± 0.004	0.082 ± 0.008	0.851 ± 0.013	0.984 ± 0.008	0.946 ± 0.013
	3	0.482 ± 0.014	0.872 ± 0.008	0.123 ± 0.008	0.809 ± 0.012	0.947 ± 0.008	0.954 ± 0.008
	4	0.498 ± 0.013	0.852 ± 0.006	0.205 ± 0.006	0.772 ± 0.012	0.879 ± 0.018	0.956 ± 0.010
	5	0.370 ± 0.013	0.825 ± 0.009	0.155 ± 0.005	0.724 ± 0.010	0.649 ± 0.020	0.956 ± 0.005
	6	0.359 ± 0.011	0.809 ± 0.006	0.263 ± 0.021	0.765 ± 0.013	0.483 ± 0.018	0.865 ± 0.018
	7	0.368 ± 0.007	0.827 ± 0.007	0.256 ± 0.012	0.758 ± 0.011	0.512 ± 0.018	0.916 ± 0.012
	8	0.380 ± 0.007	0.824 ± 0.012	0.245 ± 0.007	0.771 ± 0.013	0.555 ± 0.017	0.892 ± 0.015
	9	0.426 ± 0.012	0.836 ± 0.010	0.171 ± 0.006	0.781 ± 0.014	0.756 ± 0.023	0.907 ± 0.006
	10	0.420 ± 0.007	0.821 ± 0.016	0.141 ± 0.011	0.765 ± 0.016	0.782 ± 0.019	0.893 ± 0.017
	11	0.418 ± 0.012	0.803 ± 0.010	0.132 ± 0.010	0.719 ± 0.013	0.788 ± 0.014	0.910 ± 0.010
	12	0.506 ± 0.008	0.840 ± 0.009	0.170 ± 0.008	0.770 ± 0.017	0.943 ± 0.012	0.930 ± 0.008
LaBSE	1	0.501 ± 0.011	0.858 ± 0.008	0.170 ± 0.002	0.838 ± 0.007	0.930 ± 0.014	0.884 ± 0.010
	2	0.522 ± 0.010	0.861 ± 0.006	0.272 ± 0.003	0.817 ± 0.004	0.844 ± 0.022	0.917 ± 0.013
	3	0.535 ± 0.008	0.845 ± 0.005	0.217 ± 0.007	0.831 ± 0.010	0.946 ± 0.007	0.864 ± 0.009
	4	0.568 ± 0.008	0.849 ± 0.007	0.298 ± 0.006	0.837 ± 0.008	0.918 ± 0.014	0.865 ± 0.015
	5	0.550 ± 0.011	0.845 ± 0.007	0.254 ± 0.011	0.794 ± 0.006	0.933 ± 0.013	0.911 ± 0.015
	6	0.660 ± 0.012	0.858 ± 0.003	0.460 ± 0.021	0.816 ± 0.009	0.918 ± 0.018	0.912 ± 0.016
	7	0.595 ± 0.010	0.850 ± 0.003	0.333 ± 0.019	0.808 ± 0.006	0.934 ± 0.015	0.904 ± 0.010
	8	0.571 ± 0.013	0.848 ± 0.004	0.313 ± 0.019	0.802 ± 0.011	0.905 ± 0.018	0.906 ± 0.007
	9	0.630 ± 0.008	0.855 ± 0.004	0.380 ± 0.018	0.828 ± 0.005	0.954 ± 0.008	0.890 ± 0.007
	10	0.711 ± 0.015	0.856 ± 0.003	0.550 ± 0.018	0.835 ± 0.005	0.920 ± 0.016	0.884 ± 0.004
	11	**0.820 ± 0.009**	**0.904 ± 0.008**	**0.732 ± 0.011**	**0.873 ± 0.010**	**0.935 ± 0.013**	**0.944 ± 0.017**
	12	0.792 ± 0.013	0.883 ± 0.006	0.668 ± 0.016	0.849 ± 0.009	0.952 ± 0.013	0.928 ± 0.014
Qwen3-0.6B	1	0.549 ± 0.006	0.892 ± 0.004	0.221 ± 0.009	0.865 ± 0.008	0.972 ± 0.003	0.927 ± 0.007
	2	0.570 ± 0.013	0.882 ± 0.003	0.362 ± 0.012	0.828 ± 0.007	0.839 ± 0.028	0.951 ± 0.006
	3	0.540 ± 0.006	0.889 ± 0.005	0.214 ± 0.010	0.827 ± 0.011	0.962 ± 0.007	0.969 ± 0.011
	4	0.581 ± 0.010	0.891 ± 0.004	0.354 ± 0.008	0.844 ± 0.009	0.874 ± 0.023	0.952 ± 0.010
	5	0.486 ± 0.016	0.894 ± 0.007	0.133 ± 0.013	0.839 ± 0.011	0.943 ± 0.012	0.965 ± 0.010
	6	0.566 ± 0.011	0.897 ± 0.005	0.349 ± 0.009	0.857 ± 0.010	0.848 ± 0.020	0.949 ± 0.013
	7	0.540 ± 0.007	0.902 ± 0.006	0.211 ± 0.011	0.862 ± 0.009	0.967 ± 0.012	0.955 ± 0.015
	8	**0.608 ± 0.010**	**0.903 ± 0.006**	**0.361 ± 0.016**	**0.861 ± 0.011**	**0.927 ± 0.015**	**0.958 ± 0.009**
	9	0.610 ± 0.011	0.901 ± 0.005	0.391 ± 0.015	0.867 ± 0.009	0.892 ± 0.019	0.944 ± 0.014
	10	0.636 ± 0.012	0.890 ± 0.008	0.450 ± 0.009	0.847 ± 0.014	0.877 ± 0.020	0.946 ± 0.010
	11	0.577 ± 0.006	0.886 ± 0.005	0.257 ± 0.013	0.838 ± 0.016	0.991 ± 0.005	0.948 ± 0.015
	12	0.626 ± 0.009	0.893 ± 0.004	0.365 ± 0.019	0.844 ± 0.013	0.964 ± 0.013	0.958 ± 0.010
	13	0.638 ± 0.010	0.888 ± 0.006	0.382 ± 0.016	0.849 ± 0.013	0.970 ± 0.015	0.938 ± 0.009
	14	0.644 ± 0.010	0.885 ± 0.007	0.401 ± 0.019	0.852 ± 0.012	0.958 ± 0.015	0.929 ± 0.008
	15	0.646 ± 0.013	0.885 ± 0.005	0.409 ± 0.020	0.852 ± 0.012	0.952 ± 0.016	0.929 ± 0.017
	16	0.689 ± 0.015	0.873 ± 0.006	0.544 ± 0.017	0.837 ± 0.010	0.877 ± 0.016	0.920 ± 0.020
	17	0.657 ± 0.010	0.861 ± 0.008	0.419 ± 0.020	0.807 ± 0.014	0.964 ± 0.012	0.932 ± 0.017
	18	0.708 ± 0.011	0.868 ± 0.004	0.503 ± 0.019	0.833 ± 0.016	0.973 ± 0.005	0.913 ± 0.020
	19	0.690 ± 0.011	0.873 ± 0.005	0.462 ± 0.018	0.859 ± 0.008	0.986 ± 0.006	0.892 ± 0.015
	20	0.742 ± 0.019	0.880 ± 0.006	0.588 ± 0.023	0.858 ± 0.009	0.941 ± 0.017	0.909 ± 0.016
	21	0.757 ± 0.016	0.878 ± 0.006	0.584 ± 0.021	0.835 ± 0.009	0.981 ± 0.010	0.934 ± 0.013
	22	0.798 ± 0.014	0.882 ± 0.006	0.670 ± 0.021	0.854 ± 0.004	0.965 ± 0.006	0.917 ± 0.017
	23	0.757 ± 0.013	0.888 ± 0.007	0.578 ± 0.016	0.857 ± 0.005	0.989 ± 0.004	0.929 ± 0.016
	24	0.810 ± 0.014	0.894 ± 0.005	0.694 ± 0.022	0.861 ± 0.005	0.960 ± 0.003	0.936 ± 0.015
	25	0.732 ± 0.012	0.890 ± 0.009	0.528 ± 0.016	0.866 ± 0.005	0.997 ± 0.003	0.922 ± 0.017
	26	0.749 ± 0.013	0.895 ± 0.007	0.561 ± 0.017	0.858 ± 0.002	0.991 ± 0.003	0.943 ± 0.017
	27	0.811 ± 0.004	0.881 ± 0.003	0.741 ± 0.008	0.824 ± 0.007	0.901 ± 0.011	0.954 ± 0.011
	28	0.822 ± 0.010	0.870 ± 0.007	0.733 ± 0.014	0.800 ± 0.012	0.937 ± 0.006	0.961 ± 0.009
KaLM-mini	1	0.595 ± 0.011	0.893 ± 0.004	0.344 ± 0.002	0.846 ± 0.007	0.920 ± 0.021	0.954 ± 0.009
	2	**0.620 ± 0.017**	**0.894 ± 0.005**	**0.422 ± 0.014**	**0.845 ± 0.005**	**0.875 ± 0.021**	**0.958 ± 0.013**

(continued on next page)

Model	Layer	Task B: Top-K Ranking (appendix, 5-seed mean ± std, SIF+ABTT)					
		Acc@1 (base)	Acc@1 (SIF+ABTT)	Existing (base)	Existing (SIF+ABTT)	New (base)	New (SIF+ABTT)
	3	0.604 ± 0.010	0.894 ± 0.003	0.378 ± 0.009	0.843 ± 0.003	0.898 ± 0.017	0.960 ± 0.011
	4	0.562 ± 0.015	0.889 ± 0.004	0.265 ± 0.009	0.842 ± 0.010	0.946 ± 0.016	0.951 ± 0.011
	5	0.550 ± 0.012	0.890 ± 0.005	0.270 ± 0.007	0.841 ± 0.006	0.912 ± 0.018	0.954 ± 0.012
	6	0.522 ± 0.015	0.894 ± 0.004	0.202 ± 0.011	0.850 ± 0.006	0.935 ± 0.013	0.950 ± 0.015
	7	0.604 ± 0.017	0.894 ± 0.003	0.394 ± 0.016	0.840 ± 0.011	0.876 ± 0.020	0.963 ± 0.012
	8	0.607 ± 0.013	0.866 ± 0.003	0.387 ± 0.010	0.798 ± 0.010	0.892 ± 0.016	0.955 ± 0.010
	9	0.614 ± 0.010	0.874 ± 0.005	0.386 ± 0.009	0.821 ± 0.011	0.909 ± 0.013	0.944 ± 0.011
	10	0.602 ± 0.012	0.860 ± 0.007	0.359 ± 0.005	0.774 ± 0.012	0.917 ± 0.017	0.970 ± 0.012
	11	0.601 ± 0.012	0.864 ± 0.008	0.360 ± 0.009	0.792 ± 0.012	0.911 ± 0.020	0.957 ± 0.008
	12	0.638 ± 0.009	0.868 ± 0.006	0.415 ± 0.008	0.801 ± 0.009	0.927 ± 0.016	0.955 ± 0.005
	13	0.643 ± 0.010	0.862 ± 0.006	0.435 ± 0.007	0.792 ± 0.005	0.912 ± 0.016	0.953 ± 0.016
	14	0.653 ± 0.009	0.858 ± 0.007	0.479 ± 0.002	0.793 ± 0.009	0.879 ± 0.016	0.943 ± 0.009
	15	0.659 ± 0.007	0.868 ± 0.005	0.458 ± 0.006	0.812 ± 0.009	0.919 ± 0.010	0.941 ± 0.006
	16	0.682 ± 0.005	0.869 ± 0.008	0.475 ± 0.006	0.824 ± 0.008	0.951 ± 0.010	0.926 ± 0.009
	17	0.756 ± 0.007	0.885 ± 0.009	0.611 ± 0.018	0.841 ± 0.006	0.944 ± 0.010	0.942 ± 0.015
	18	0.780 ± 0.010	0.881 ± 0.007	0.635 ± 0.016	0.840 ± 0.002	0.967 ± 0.007	0.933 ± 0.016
	19	0.804 ± 0.013	0.876 ± 0.007	0.672 ± 0.022	0.831 ± 0.007	0.977 ± 0.003	0.935 ± 0.008
	20	0.793 ± 0.011	0.877 ± 0.004	0.649 ± 0.020	0.831 ± 0.003	0.979 ± 0.004	0.936 ± 0.008
	21	0.804 ± 0.011	0.871 ± 0.007	0.672 ± 0.018	0.830 ± 0.005	0.975 ± 0.006	0.924 ± 0.012
	22	0.870 ± 0.008	0.873 ± 0.004	0.810 ± 0.007	0.812 ± 0.005	0.948 ± 0.008	0.952 ± 0.009
	23	0.873 ± 0.007	0.884 ± 0.004	0.811 ± 0.008	0.833 ± 0.004	0.953 ± 0.007	0.951 ± 0.007
	24	0.848 ± 0.011	0.856 ± 0.005	0.749 ± 0.019	0.806 ± 0.009	0.975 ± 0.007	0.921 ± 0.010

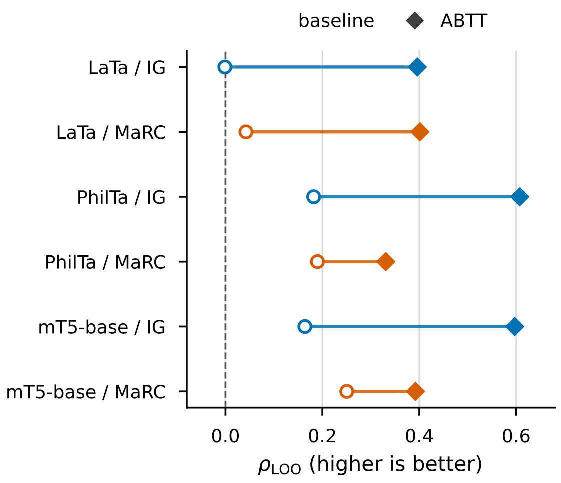

Figure 4: ρ_{LOO} **foreground view for the main candidate-attribution result.** Each line connects the baseline and ABTT variants for one model-method cell at the predeclared operational attribution layer. ABTT improves the leave-one-out rank-correlation signal in all six cells; Table 5 shows the ERASER-style disagreements which qualify this narrower faithfulness claim.

F Attribution Metric Sweeps

Figure 4 plots the rank-faithfulness column of Table 5. The rest of this appendix expands the main attribution table across Sufficiency and Comprehensiveness thresholds of 10%, 25%, and 50%, and MinFrac recovery thresholds of 0.70, 0.80, 0.90, and 0.95. The main text uses one global midpoint setting rather than selecting thresholds per model or per method. Table 17 gives the sweep for integrated gradients and retrieval-adapted MaRC, and Table 18 gives it for the supplemental attribution methods.

G Qualitative Attribution Examples

This appendix keeps the example-level views for one PhilTa retrieval pair. Figure 5 gives the integrated-gradient pair matrix before and after ABTT. The attention map in Figure 6 is context only, since ABTT is applied after encoding. The retrieval-adapted MaRC masks in Figure 7 show the learned-mask view for the same pair.

H Cluster Geometry

Figures 8 and 9 give the t-SNE and UMAP panels for LaTa, PhilTa, and mT5-base, and Figures 10 and 11 extend them to LaBSE, Qwen3-0.6B, and KaLM-mini. All four use the same six directories and the same baseline versus pure ABTT comparison.

Model	Method	$\rho_{\text{LOO}} \uparrow$		Suff@10% ↑		Suff@25% ↑		Suff@50% ↑		Comp@10% ↑		Comp@25% ↑		Comp@50% ↑		MinFrac@0.70 ↓		MinFrac@0.80 ↓		MinFrac@0.90 ↓		MinFrac@0.95 ↓	
		base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt
LaTa	IG	-0.001	0.396	0.967	0.428	0.976	0.681	0.975	0.921	0.320	0.267	0.719	0.581	0.850	0.884	0.040	0.293	0.041	0.379	0.045	0.489	0.047	0.557
	MaRC	0.042	0.401	0.404	0.475	0.534	0.645	0.582	0.863	0.116	0.250	0.361	0.539	0.506	0.845	0.368	0.356	0.383	0.459	0.396	0.568	0.438	0.624
PhilTa	IG	0.182	0.607	0.816	0.848	0.959	0.963	0.988	1.057	0.704	0.436	0.969	0.747	1.010	0.992	0.027	0.090	0.079	0.141	0.162	0.229	0.215	0.300
	MaRC	0.190	0.331	0.786	0.531	0.911	0.754	0.949	0.935	0.657	0.237	0.963	0.521	0.978	0.793	0.058	0.281	0.104	0.358	0.221	0.448	0.327	0.503
mT5-base	IG	0.164	0.597	0.810	0.785	0.892	1.027	0.937	1.192	0.054	0.376	0.110	0.717	0.284	1.071	0.055	0.161	0.116	0.214	0.338	0.291	0.623	0.336
	MaRC	0.250	0.392	0.865	0.597	0.913	0.827	0.951	1.055	0.010	0.279	0.042	0.592	0.118	0.923	0.026	0.277	0.066	0.346	0.212	0.434	0.416	0.488

Table 17: Appendix attribution sweep for the two paper-facing attribution views, IG and retrieval-adapted MARC, across the three main-paper models. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at $\{10\%, 25\%, 50\%\}$; MinFrac is the minimum retained query-token fraction needed to recover $\tau \in \{0.70, 0.80, 0.90, 0.95\}$ of the full cosine (lower is better). Ratio metrics require $|S_v(q_{\text{full}}, c)| \geq 0.05$; ρ_{LOO} excludes $|\Delta| < 10^{-6}$. The completeness report contains no missing metric cells.

Model	Method	$\rho_{\text{LOO}} \uparrow$		Suff@10% ↑		Suff@25% ↑		Suff@50% ↑		Comp@10% ↑		Comp@25% ↑		Comp@50% ↑		MinFrac@0.70 ↓		MinFrac@0.80 ↓		MinFrac@0.90 ↓		MinFrac@0.95 ↓	
		base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt	base	abtt
LaTa	BERTScore	0.204	0.430	1.035	0.365	1.047	0.626	1.047	0.904	0.916	0.221	1.301	0.436	1.280	0.719	0.018	0.318	0.021	0.394	0.026	0.481	0.036	0.534
	OT	-0.001	0.396	0.967	0.428	0.976	0.681	0.975	0.921	0.320	0.267	0.719	0.581	0.850	0.884	0.040	0.293	0.041	0.379	0.045	0.489	0.047	0.557
	Att-weighted	-0.137	0.106	-0.575	0.142	-0.377	0.295	-0.131	0.588	0.001	0.159	-0.005	0.330	-0.019	0.543	0.746	0.569	0.756	0.653	0.774	0.747	0.796	0.804
	Att-standalone	0.005	0.368	0.115	0.358	0.239	0.640	0.409	0.873	0.043	0.251	0.112	0.536	0.342	0.830	0.516	0.333	0.531	0.422	0.558	0.537	0.584	0.597
	DLA	0.054	0.278	1.017	0.350	1.004	0.607	1.003	0.820	0.937	0.239	1.289	0.505	1.248	0.792	0.018	0.368	0.021	0.471	0.026	0.583	0.036	0.650
	random	0.003	-0.003	0.403	0.222	0.726	0.360	0.906	0.599	0.024	0.072	0.046	0.181	0.105	0.393	0.176	0.561	0.182	0.667	0.192	0.779	0.212	0.842
	inverse	0.001	-0.396	-0.046	-0.014	0.122	-0.001	0.151	0.124	0.000	-0.046	0.001	-0.035	0.025	0.086	0.553	0.837	0.562	0.882	0.602	0.920	0.629	0.942
PhilTa	BERTScore	0.309	0.550	0.861	0.771	0.985	0.942	1.035	1.050	0.687	0.313	1.032	0.580	1.042	0.808	0.026	0.128	0.070	0.188	0.190	0.284	0.280	0.343
	OT	0.182	0.607	0.816	0.848	0.959	0.963	0.988	1.057	0.704	0.436	0.969	0.747	1.010	0.992	0.027	0.090	0.079	0.141	0.162	0.229	0.215	0.300
	Att-weighted	0.053	0.108	0.060	0.341	0.163	0.584	0.219	0.753	0.011	0.157	0.028	0.326	0.059	0.524	0.673	0.397	0.693	0.505	0.742	0.627	0.795	0.698
	Att-standalone	0.046	0.205	0.017	0.351	0.038	0.654	0.064	0.827	0.007	0.163	0.040	0.378	0.069	0.616	0.773	0.323	0.782	0.417	0.796	0.537	0.808	0.612
	DLA	0.203	0.487	0.805	0.863	0.942	0.910	0.991	0.957	0.715	0.438	1.017	0.729	1.028	0.927	0.028	0.087	0.080	0.142	0.194	0.260	0.260	0.346
	random	-0.001	-0.010	0.589	0.288	0.826	0.478	0.953	0.709	0.011	0.050	0.016	0.124	0.066	0.288	0.135	0.437	0.145	0.563	0.204	0.713	0.284	0.798
	inverse	-0.182	-0.607	-0.009	-0.074	-0.025	-0.074	-0.009	0.014	-0.001	-0.049	-0.001	-0.073	0.013	-0.055	0.917	0.920	0.921	0.950	0.932	0.975	0.942	0.987
mT5-base	BERTScore	0.271	0.556	0.615	0.694	0.776	1.005	0.917	1.197	0.014	0.270	0.038	0.579	0.082	0.906	0.148	0.199	0.214	0.255	0.333	0.322	0.452	0.358
	OT	0.164	0.597	0.810	0.785	0.892	1.027	0.937	1.192	0.054	0.376	0.110	0.717	0.284	1.071	0.055	0.161	0.116	0.214	0.338	0.291	0.623	0.336
	Att-weighted	0.173	0.523	0.872	0.890	0.923	1.008	0.942	1.099	0.012	0.364	0.048	0.687	0.249	0.986	0.036	0.117	0.074	0.181	0.239	0.278	0.526	0.346
	Att-standalone	0.110	0.064	0.640	0.180	0.650	0.455	0.862	0.643	0.034	0.184	0.068	0.362	0.135	0.464	0.066	0.471	0.137	0.575	0.412	0.718	0.608	0.786
	DLA	0.129	0.481	0.296	0.922	0.786	0.996	0.949	1.010	0.029	0.408	0.045	0.683	0.083	0.908	0.211	0.097	0.259	0.159	0.347	0.257	0.460	0.337
	random	-0.000	-0.004	0.846	0.296	0.926	0.483	0.969	0.687	0.004	0.059	0.013	0.153	0.033	0.317	0.045	0.460	0.079	0.591	0.178	0.735	0.313	0.813
	inverse	-0.164	-0.597	-0.127	-0.235	0.203	-0.237	0.722	-0.065	0.022	-0.108	0.045	-0.187	0.064	-0.189	0.472	0.896	0.563	0.936	0.740	0.971	0.863	0.985

Table 18: Supplemental attribution sweep for additional methods and sanity baselines. *random* and *inverse* are diagnostic baselines, not paper-facing attribution methods. Direction arrows are printed in the column headers. Sufficiency and Comprehensiveness are ratio metrics at $\{10\%, 25\%, 50\%\}$; MinFrac is the minimum retained query-token fraction needed to recover $\tau \in \{0.70, 0.80, 0.90, 0.95\}$ of the full cosine (lower is better). Ratio metrics require $|S_v(q_{\text{full}}, c)| \geq 0.05$; ρ_{LOO} excludes $|\Delta| < 10^{-6}$. The completeness report contains no missing metric cells.

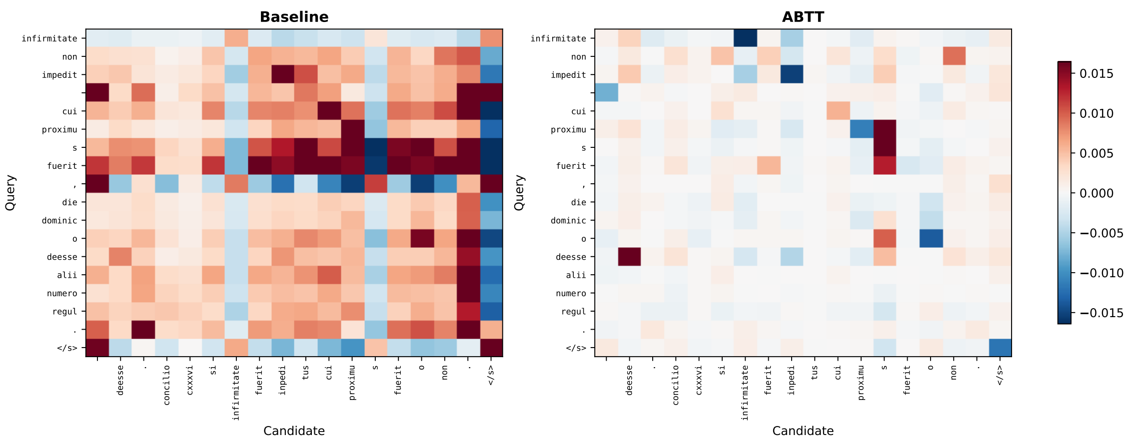

Figure 5: Derived pairwise heatmap for one PhilTa retrieval case. The left panel shows the baseline representation and the right panel shows the same example after ABTT. Both axes use original decoded tokens. Cell values combine token-token cosine similarity with single-sequence integrated-gradient scores.

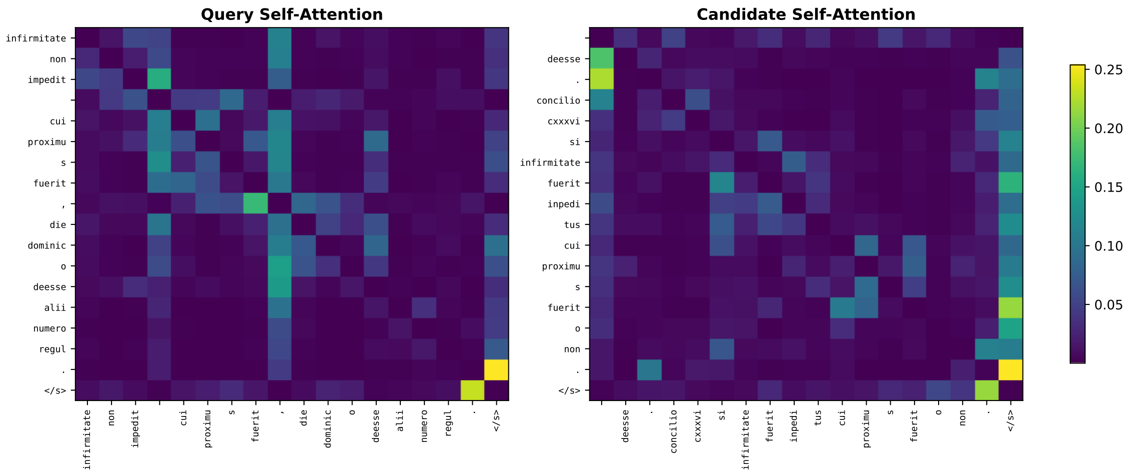

Figure 6: Companion self-attention maps for the same PhilTa example as Figure 5. ABTT is applied after encoding, so attention is included only as supporting context.

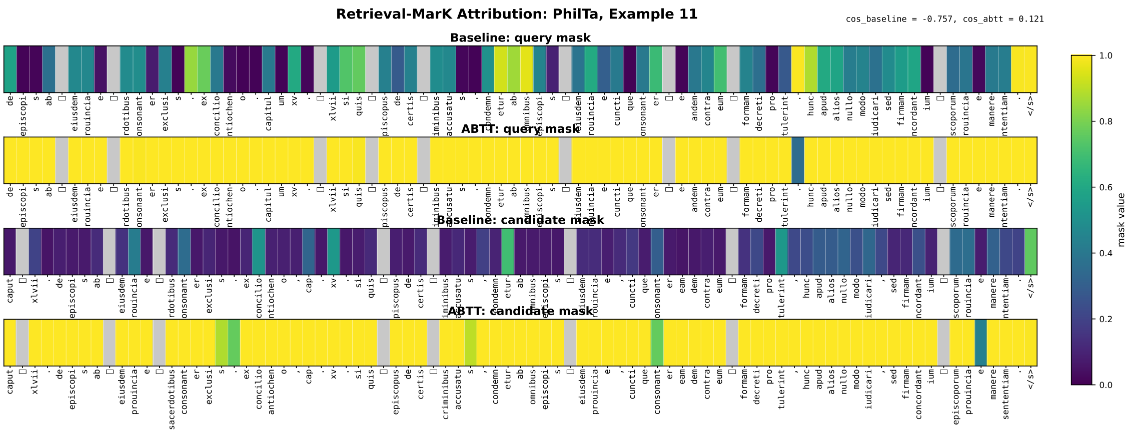

Figure 7: Retrieval-adapted MaRC masks for the same PhilTa pair as Figure 5. Top row: mask on the query with the candidate fixed. Bottom row: mask on the candidate with the query fixed. Left column: baseline cosine target. Right column: ABTT-corrected cosine target.

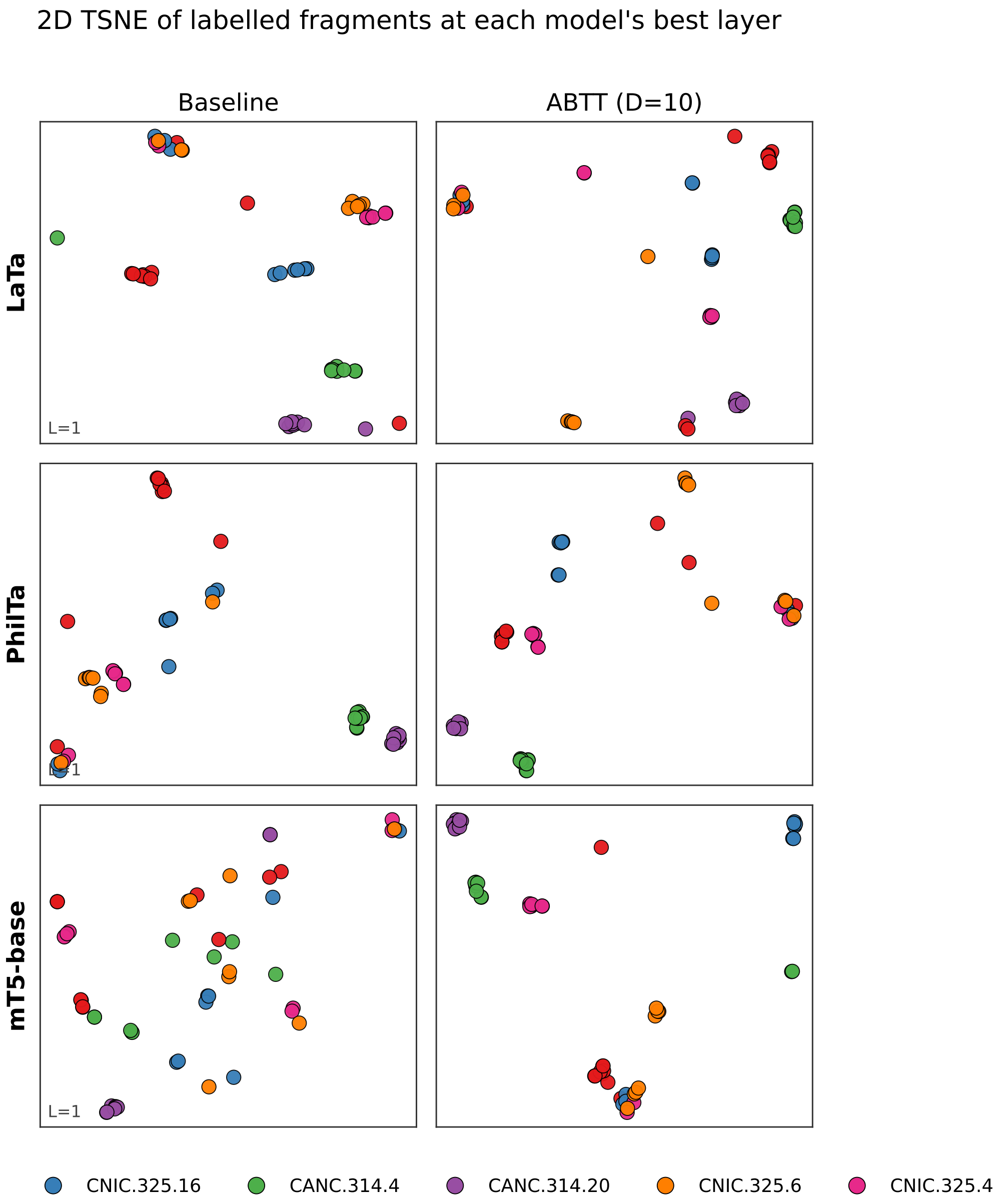

Figure 8: t-SNE projections of the 1,705 labelled fragments at each model's selected layer for LaTa, PhilTa, and mT5-base. Columns compare baseline mean pooling with pure ABTT using $D = 10$. Colors mark the six most-populated directories.

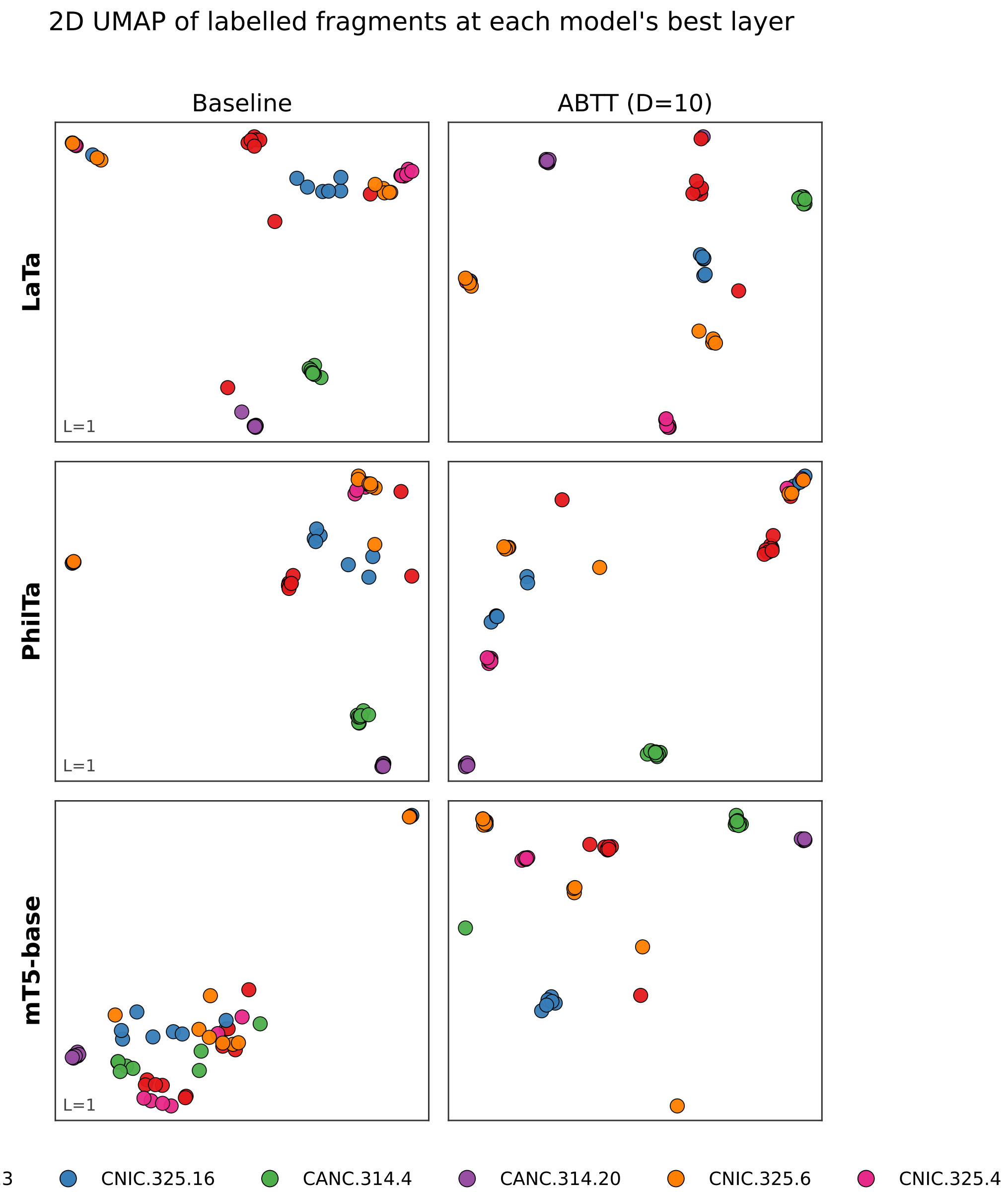

Figure 9: UMAP projections with the same configuration as Figure 8. The same cluster-tightening pattern appears under a different projection method.

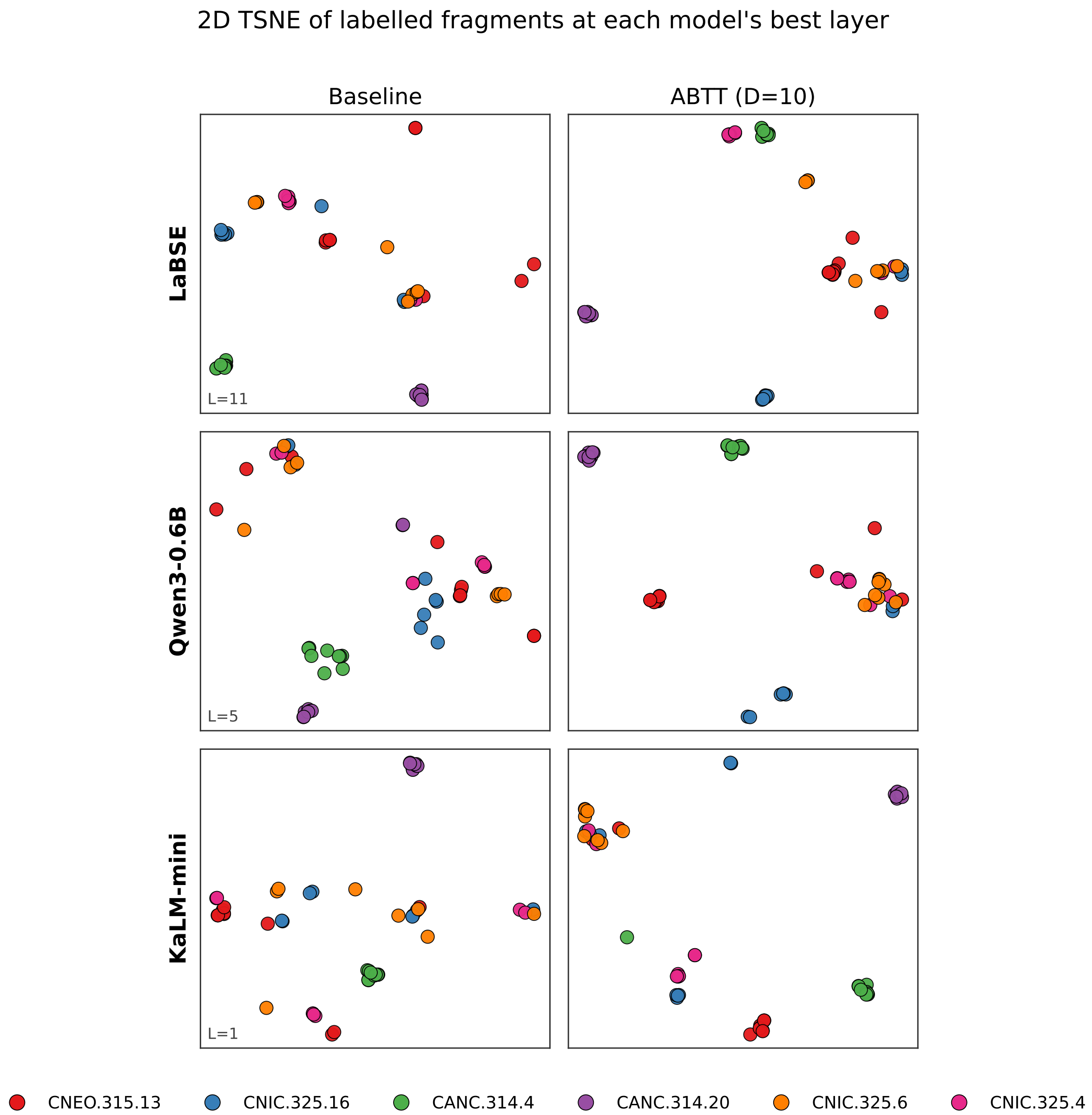

Figure 10: t-SNE projections of the six most-populated directories at each selected layer for LaBSE, Qwen3-0.6B, and KaLM-mini. Columns compare baseline mean pooling with pure ABTT using $D = 10$.

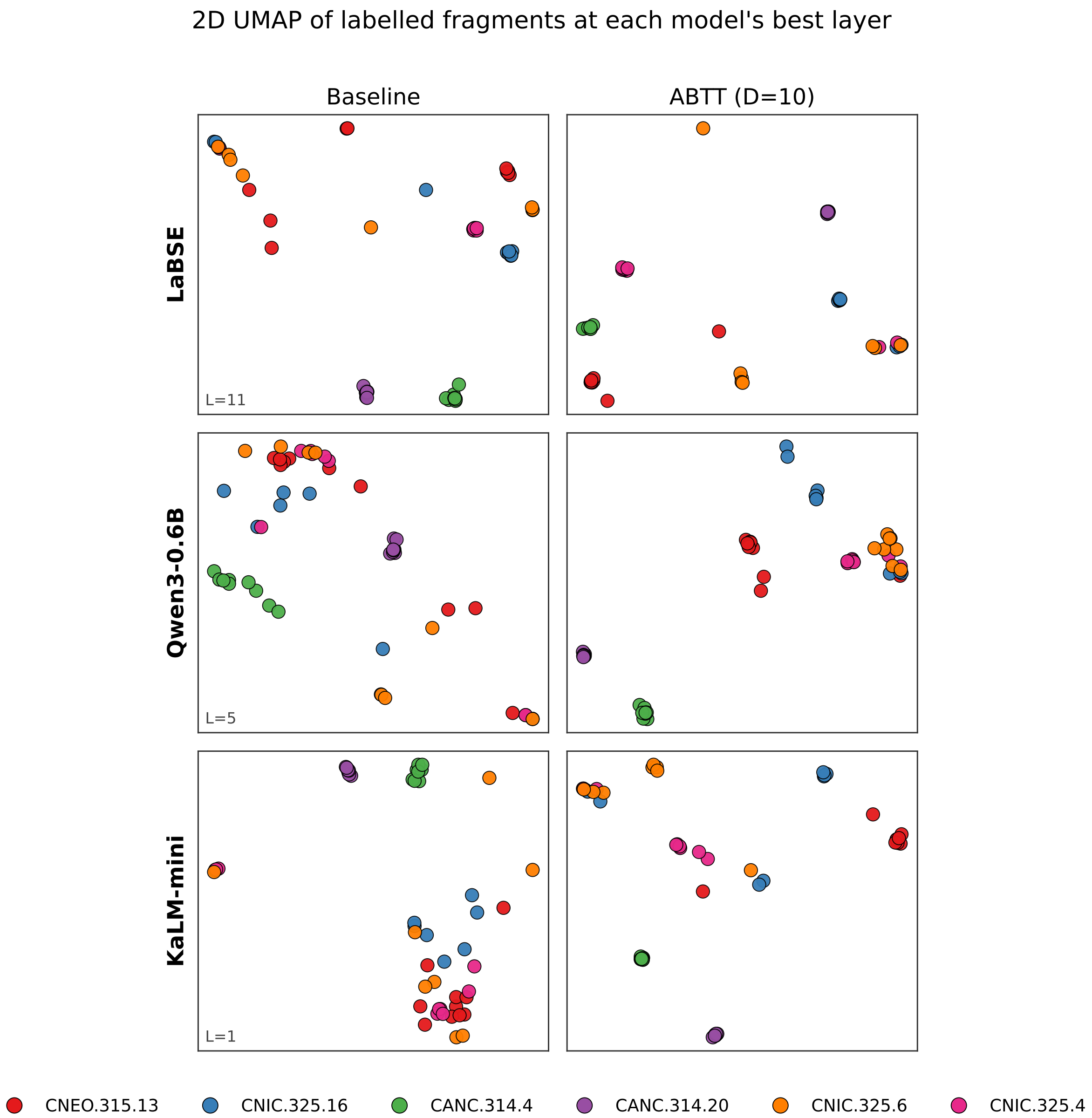

Figure 11: UMAP projections matching Figure 10. The cluster-tightening pattern appears under a second projection method.